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Course of Spacecraft Attitude Dynamics

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Project Specifications

	Assigned specification	Modifications	Motivations
Platform	Microsat	-	-
Attitude parameters	Quaternions	-	-
Mandatory sensor	Gyroscopes	Sun sensor and horizon sensor added	See paragraph 1.4
Actuators	CMG	Four thrusters	See paragraph 1.4

Abstract

This project presents the design, modeling, and validation of an Attitude Determination and Control System (ADCS) for a microsatellite operating in a Sun-synchronous low Earth orbit, inspired by high-resolution Earth observation missions such as PlanetLabs SkySat. The primary mission objective is to achieve precise nadir-pointing with full attitude control in the Local Vertical Local Horizontal (LVLH) reference frame to support imaging payload requirements.

A comprehensive simulation environment was developed, integrating spacecraft rigid-body dynamics, quaternion-based kinematics, orbital mechanics, and environmental disturbances, including gravity gradient and geomagnetic torques. The attitude determination architecture combines gyroscope measurements with Sun and horizon sensors, using the Q-method algorithm to ensure robust and accurate attitude estimation while mitigating sensor drift.

The control architecture is structured into three sequential operational phases: de-tumbling, slew maneuver, and nominal pointing. Initial angular rate reduction is achieved through a thruster-based de-tumbling strategy. Following that, a nonlinear quaternion feedback control law for large-angle reorientation is implemented using a pyramid-configured Control Moment Gyroscope (CMG) cluster, with singularity avoidance ensured through null-motion control. Finally, during nominal operations, a Linear Quadratic Regulator (LQR) is employed to maintain precise LVLH tracking.

Simulation results demonstrate rapid de-tumbling within 30 seconds, successful slew maneuver execution with bounded angular rates, and stable nadir-pointing performance. A Monte Carlo campaign with randomized initial conditions confirms global stability and robustness, with maximum observed pointing errors remaining an order of magnitude below allowable limits.

Overall, the proposed ADCS architecture satisfies mission requirements and demonstrates strong robustness and control authority, providing a solid foundation for future development.
Keywords: ADCS, Microsatellite, Nadir Pointing, Quaternion Control, CMG, Simulation, Spacecraft Attitude Dynamics

Contents

1. Introduction

1.1 Mission Objectives and Requirements

The goal of this project was to investigate and design the ADCS (Attitude Determination and Control System) of a microsat-class satellite. In parallel with it, a simulation environment, containing dynamics of the spacecraft, perturbation torques and basic orbital mechanics model was developed.
The inspiration of the mission was similarly-sized *PlanetLabs SkySat* satellite used for high-resolution optical Earth observation [5]. Since it creates a multi-frame image stacking, a precise nadir pointing architecture needed to be delivered.

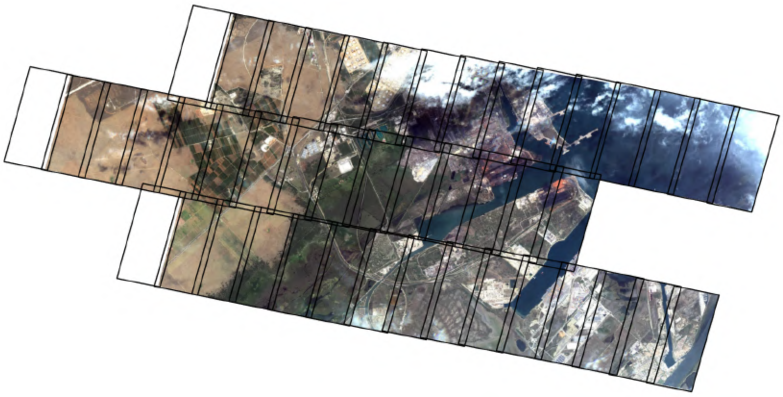

Figure 1.1: Sample images taken by SkySat [4]

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1.2 Orbit Description and Parameters

The orbit is Sun-synchronous

$\mathbf{a}\,[Km]$	$\mathbf{e}\,[-]$	$\mathbf{i}\,[deg]$	$\mathbf{\Omega}\,[deg]$	$\omega\,[deg]$	$\mathbf{T}\,[s]$
6793.635	0.0007	96.99	193.4991	218.1753	5572.68

Table 1.1: Orbital Parameters

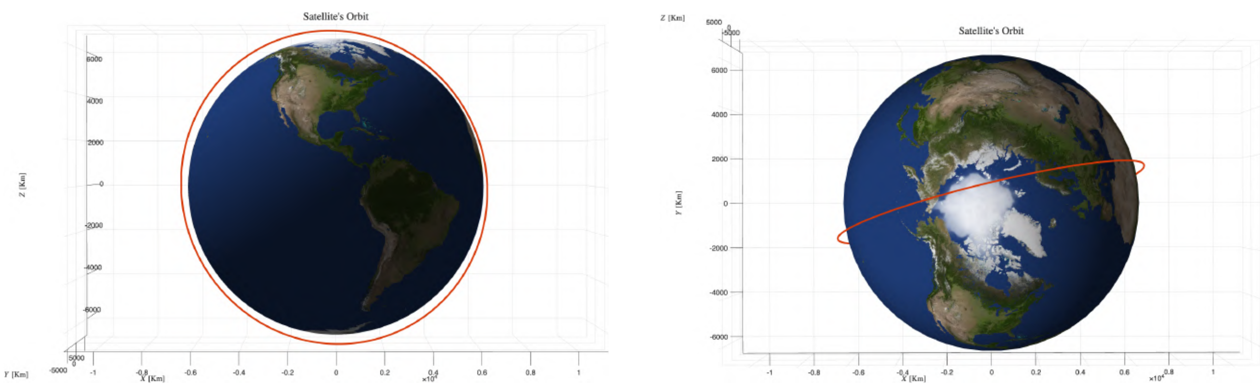


Figure 1.2: Satellite's Orbit

1.3 Spacecraft Geometry and Inertial properties

Parameter	Value
Mass	100 kg
Height (z)	1.2 m
Width (x)	0.5 m
Depth (y)	0.5 m
I_x	14.0833 kg m^2
I_y	14.0833 kg m^2
I_z	4.0167 kg m^2

Table 1.2: Dimensional parameters of the satellite

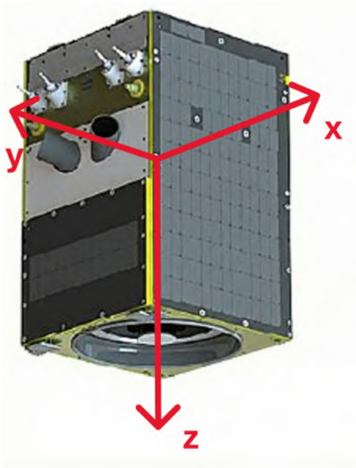


Figure 1.3: Image of the satellite

1.4 ADCS Architecture

Mission architecture can be split into three basic steps:

1. De-tumbling
2. Slew maneuver into nadir direction
3. Nadir direction and orientation tracking

As an example, the LVLH (local vertical, local horizontal) frame was selected as the desired attitude to simulate the need for precise imaging with a high–aspect ratio camera sensor - only nadir direction tracking would be insufficient. The following definition of LVLH was used:

$$\hat{x}_{\text{LVLH}} = \hat{y} \times \hat{z} \qquad \hat{y}_{\text{LVLH}} = -\frac{\mathbf{h}_{\text{ECI}}}{||\mathbf{h}_{\text{ECI}}||} \qquad \hat{z}_{\text{LVLH}} = -\frac{\mathbf{r}_{\text{ECI}}}{||\mathbf{r}_{\text{ECI}}||} \tag{1.1}$$

To meet these requirements, a set of additional components was incorporated.

Additional sensors justification
In the initial configuration, the spacecraft was equipped with only gyroscopic sensor. However, in order to define the attitude with respect to Earth, a minimum one additional sensor measuring external environment was needed. The mission objective, required a full attitude determination, since only nadir direction tracking would not satisfy the correct camera orientation requirement. As a result, sun and horizon sensors were added. It was perceived as a cheaper and more robust alternative to the star tracker. This combination ensures that two independent, non-collinear vectors (Sun vector and Earth vector) are available for attitude determination algorithms throughout the orbit, balancing precision and robustness. Considering the high inclination of the spacecraft orbit, the magnetometer may provide less accurate attitude information, as the Earth's magnetic field lines become more vertical and less distinct in direction at high latitudes. This reduces the observability and precision of angle measurements.

Additional actuator justification
In the primary setup, the satellite was equipped with a CMG cluster, which provides high levels of torque, excellent for dynamic attitude control. However, as any actuator based on the exchange of angular momentum with the spacecraft, it has a limit of possible momentum storage. It was

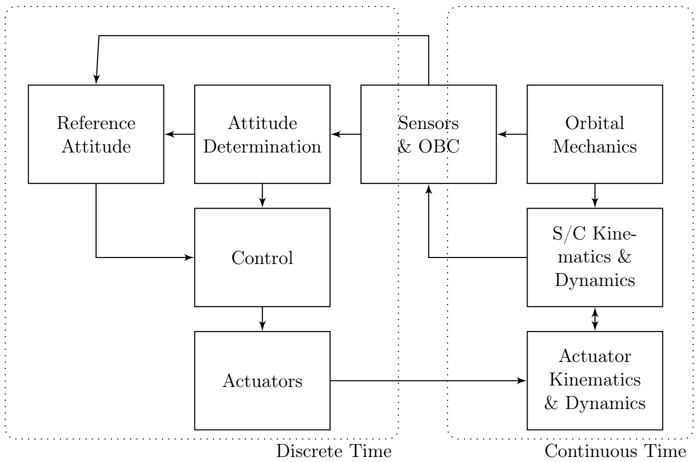
observed (Chapter 2.5.1 - 2.5.2) that CMG is insufficient for de-tumbling maneuver under medium to high initial angular velocities. Based on the *SkySat* configuration, a set of four thrusters was added, which in the simulation were activated exclusively during de-tumbling phase. In reality, their application extends to orbital maneuvers and CMG de-saturation.

2. Environment & Spacecraft Modeling

2.1 Simulation Architecture

Simulation architecture was split into three basic sections:

- continuous time representing real environment,
- discrete time representing AOCS software,
- transition from continuous to discrete performed by sensors and on-board computer.



2.2 Dynamics and Kinematics

Dynamic model of the spacecraft was represented with the extension of Euler's equation

$$\mathbf{I}\dot{\omega} + \omega \times \mathbf{I}\omega + \dot{\mathbf{A}}\mathbf{h_r} + \omega \times \mathbf{A}\mathbf{h_r} = \mathbf{M_{disturb}}, \tag{2.1}$$

where it was assumed that control moment gyroscope has a constant speed and it's moment of inertia is negligible in comparison to the satellite's, therefore $I_{satellite} = const.$ Values of angular speed ω are around principal axis and moment of inertia is represented by the diagonal matrix.

Quaternion formulation was used to describe the kinematics of the satellite

$$\dot{\mathbf{q}} = \frac{1}{2}\mathbf{\Omega}\mathbf{q}(t), \quad \mathbf{\Omega} = \begin{bmatrix} 0 & \omega_w & -\omega_v & \omega_w \\ -\omega_w & 0 & \omega_u & \omega_v \\ \omega_v & -\omega_u & 0 & \omega_w \\ -\omega_u & -\omega_v & -\omega_w & 0 \end{bmatrix}, \tag{2.2}$$

where ω are the rotational velocities of the body with respect to the inertial frame represented in body principal axis. The used inertial frame was the Earth-centered inertial frame (ECI).

2.3 Environmental Disturbance Modeling

The LEO orbit environment is characterized by four primary perturbations: gravity gradient, magnetic field, solar radiation pressure and aerodynamic drag. The torques were first estimated with the mathematical formulas listed in the table. While aerodynamic drag constitutes a major disturbance component, it was excluded from this simulation due to the high uncertainty associated with atmospheric density fluctuations. Consequently, only gravity gradient and magnetic field torques were modeled in the simulation.

Dist. torque	**Order of magnitude**
Gravity grad.	$T_{max} = \frac{3Gm_t}{2R^3}\lvert I_M - I_m\rvert \sim 10^{-5}$
Solar rad. press.	$T_{max} = P_s A_s (1+q)(c_{ps} - c_g) \sim 10^{-7}$
Mag. field	$T_{max} = D_s B_{max} \sim 10^{-6}$
Aerodyn.	$T_{max} = \frac{1}{2}\rho V^2 A_s C_D (c_{pa} - c_g) \sim 10^{-6}$

The used values for the estimation were:

- I_M = 14.0833 kg m^2
- I_m = 4.0167 kg m^2
- μ_{earth} = 3.986 × 10^{14} m^3/s^2
- R = R$_e$ + h ≃ 6.778 × 10^6 m
- Solar press. P$_s$ = 4.56 × 10^{-6} Pa
- Exposed area A$_s$ = 0.5 × 1.2 = 0.6 m^2
- Reflectance (assumed) q ≃ 0.6
- Sol. press. moment arm $c_{ps} - c_g$ = 0.05 m
- Residual dipole D$_s$ = |**m**| = 0.052 Am^2 with **m** = [0.01 0.05 0.01]^T Am^2
- B$_{max}$ = 4 × 10^{-5} T at 400 km
- ρ ≃ 10^{-11} kg/m^3 at 400 km
- C$_D$ = 2.2
- Aer. moment arm $c_{pa} - c_g$ = 0.02 m
- V = $\sqrt{\mu/R}$ ≃ 7670 m/s

2.3.1 Gravity Gradient

The gravity gradient torque arises from the interaction between the non-uniform gravitational field and the satellite's mass distribution. This torque acts to align the spacecraft's axis of minimum inertia with the nadir vector:

$$T_{gg} = \frac{3Gm_t}{R^3}\begin{Bmatrix}(I_z - I_y)c_2c_3\\(I_x - I_z)c_1c_3\\(I_y - I_x)c_1c_2\end{Bmatrix} = -\frac{3\mu}{R^3}\cdot[\mathbf{Ic}\times\mathbf{c}]$$

where $\mathbf{c} = \mathbf{A}_{B|L}[1,0,0]^T$. Assuming the orbital is almost circular and calling $n^2 = \frac{Gm_t}{R^3}$ it is possible to rewrite the formula as:

$$T_{gg} = 3n^2\begin{Bmatrix}(I_z - I_y)c_2c_3\\(I_x - I_z)c_1c_3\\(I_y - I_x)c_1c_2\end{Bmatrix}$$

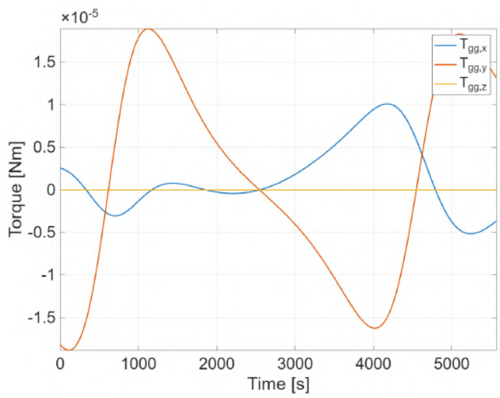


Figure 2.1: Gravity gradient torque acting on an uncontrolled S/C with zero initial condition

2.3.2 Magnetic Field

The Earth's magnetic field was modeled using the 13th-order, 14th-generation IGRF (International Geomagnetic Reference Field) model. Model verification was performed by comparing the calculated magnetic field magnitude at an altitude of 500 km (Figure 2.2) with publicly available reference data .

The torque due to the Earth's magnetic field is given by

$$\mathbf{T} = \mathbf{m} \times \mathbf{B}, \tag{2.3}$$

where $\mathbf{m} = [0.01\ 0.05\ 0.01]^T$ is the geomagnetic field vector at the satellite's position, and $\mathbf{m}$ is the spacecraft's net magnetic dipole moment resulting from onboard electrical currents and materials.

The value of $\mathbf{m}$ was selected according to [1], assuming *Class 3* spacecraft properties, representing a worst-case disturbance torque scenario.

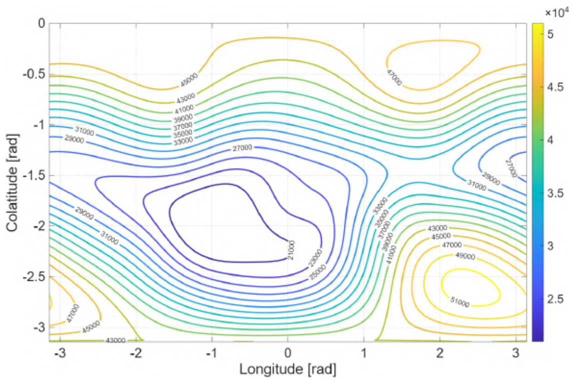


Figure 2.2: Total magnetic field intensity [nT] at 500km

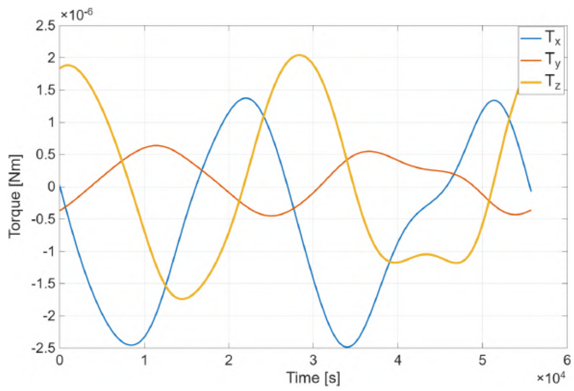


Figure 2.3: Magnetic filed torque acting on an uncontrolled S/C with zero initial conditions

2.4 Sensor Modeling

2.4.1 Gyroscope

To measure the angular velocity of the satellite around the three body axes, a three axis gyroscopes assembly was used. The sensors were assumed to be calibrated both prior to launch and after the de-tumbling phase. By applying the kinematic model described in 2.2, the attitude quaternion was propagated from these rate measurements. However, despite calibration, residual errors such as bias instability and angular random walk lead to a drift from the true state over time. To compensate these errors, a reset logic utilizing the $\boldsymbol{\omega}_I$ measurements (body angular rate with respect to the inertial frame) from the sun and horizon sensor was implemented and triggered every 300 seconds. The model of the sensor considers latency, misalignment errors, linear and non-linear scale factor errors, constant bias and bias instability, white noise, quantization, saturation and discretization.

Figure 2.4: Epson M-G370PDT0 IMU

Product parameter	Value
Bias instability	$0.8°/h$
ARW	$0.03\ °/\sqrt{h}$
nBit	32
Range	$\pm 200\ °/s$
Frequency	10 Hz
Mounting & use	**Value**
Bias after calibration	$1°/h$

Table 2.1: Selected IMU specification from [8]

2.4.2 Sun sensor

Sun sensor measures the direction of the sun with respect to the body frame. To verify the robustness of the control laws continuously throughout the orbital period, the simulation assumes a simplified visibility model where the Sun is treated as always available.

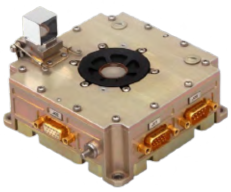


Figure 2.5: Leonardo S3 Sun sensor

Product parameter	Value
Accuracy	$< 0.02°(2\sigma)$
Resolution	$< 0.005°$
FOV	$120° \times 120°$ (resizable)
Mounting & use	**Value**
Mounting side	Z

Table 2.2: Selected sun sensor specification from [7]

2.4.3 Horizon sensor

The horizon sensor is crucial for the mission, as precise nadir pointing is required for the imaging payload. The instrument determines the direction of the Earth's center relative to the spacecraft's body frame. In the model, the ideal Earth's direction vector $\mathbf{r}_b$ was utilized as the baseline to derive the roll and pitch angles between it and the body frame axis. To ensure high fidelity, the sensor was modeled taking into account misalignment errors, scale factor errors, white noise, constant bias, quantization and discretization. The parameters were mainly taken from the datasheet.

Figure 2.6: Leonardo IRES horizon sensor

Product parameter	Value
Accuracy	$< 0.05°(3\sigma)$
Bias	$< 0.02°$
FOV	$44° \times 26°$ (resizable)
Mounting & use	**Value**
Mounting side	Z

Table 2.3: Selected horizon sensor specification from [6]

2.5 Actuator Modeling

2.5.1 Control moment gyroscope

A pyramid configuration containing four single-axis, constant speed CMG's was chosen. For the microsat class, usage of this actuator is still uncommon, however its implementation allows for high agility maneuvers. Axis configuration is shown in (Figure 2.7), where g is gimbal axis and s is in the direction of angular momentum. The control torque is controlled by the appopriate angular velocities of all gimbal angles δ using the following equation

$$\dot{\delta} = -\mathbf{B}^{*}[\mathbf{M_c} + \omega_I \times \mathbf{A}\mathbf{h_r}] \tag{2.4}$$

where ω_I is the measured angular velocity of the spacecraft, $\mathbf{h_r}$ is the constant angular velocity vector of each gimbal, $\mathbf{M_c}$ is the controller torque request and $\mathbf{A}$ and $\mathbf{B}$ matrices are defined below.

$$\mathbf{A}_{:,i} = \mathbf{R_z}((i-1)\frac{\pi}{2})\mathbf{R_y}(\beta)\mathbf{R_z}(\delta_i) \cdot [\mathbf{0}, \mathbf{h_r}, \mathbf{0}]^{\mathbf{T}} \quad i = 1, 2, 3, 4, \tag{2.5}$$

$$\mathbf{B}_{:,i} = \mathbf{R_z}((i-1)\frac{\pi}{2})\mathbf{R_y}(\beta)\mathbf{R_z}(\delta_i)\mathbf{S_z} \cdot [\mathbf{0}, \mathbf{h_r}, \mathbf{0}]^{\mathbf{T}} \quad i = 1, 2, 3, 4, \tag{2.6}$$

$$\mathbf{S_z} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \tag{2.7}$$

A CMG cluster can encounter a singular position - it is unable to provide a accelerating torque in a particular direction. To mitigate that singularity, a null motion defined as

$$\mathbf{B}\dot{\sigma}_{null}k = \mathbf{0}, \tag{2.8}$$

where k is a constant gain, added to the prescribed gimbal angular velocity. As a result, CMG gimbals will not get stuck in the singular position. In the simulation, the sign of k constant was further altered real-time to provide a smooth motion request.

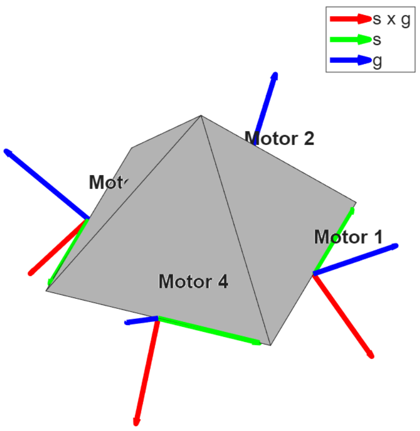


Figure 2.7: CMG axis definitions

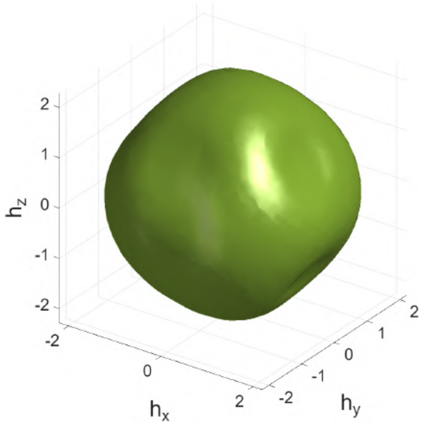


Figure 2.8: Surface of maximum available angular momentum [Nm]

Finally, since the control moment gyroscope does not exchange the momentum with the environment, under certain conditions it reaches a saturation point in a particular direction. It is represented by the surface of CMG's maximum angular momentum storage 2.8. It is particularly significant during de-tumbling maneuver and when satellite is subjected to a constant external torque.

Figure 2.9: Veoware microCMG

Product parameter	Value
Momentum	0.7 Nms
Torque (max)	1.1 Nm
Mass	2.4 kg
Mounting & use	**Value**
Pyramid slope angle	54.73

Table 2.4: Sinigle CMG specification

2.5.2 Thrusters

Four thrusters were added to the model in order to provide a robust de-tumbling capabilities for high initial angular velocities, where S/C angular momentum is lying close or outside of the surface in (Figure 2.8), eg $\omega_{y0} > 8\frac{deg}{s}$. Thrusters were spaced on one of the satellite panels in a configuration shown below. Their reaction time to the request was modeled as instantaneous, which is a valid simplification considering their specification and low OBC frequency used in the simulation.

Figure 2.10: Liftero 1N Class thruster

Product parameter	Value
Thrust range	1.10 - 2.26N
Specific impulse	269 - 292 s
Minimum impulse bit	12 mNs (cold-gas)
Mounting & use	**Value**
Thruster angle α	30 °
x_{thruster}	0.167 m
z_{thruster}	0.4 m

Table 2.5: Sinigle thruster specification

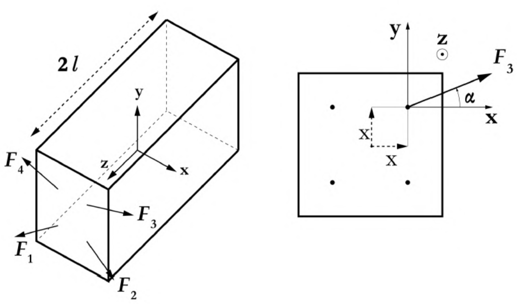


Figure 2.11: Thruster position and orientation [2]

3. Attitude Determination and Control

3.1 Attitude Determination Algorithms

The Q-method algorithm was implemented to estimate the spacecraft's attitude. The algorithm computes the weighted matrix **B** utilizing the body-frame direction vectors from the Sun and Horizon sensors, their corresponding inertial references, and a third synthetic measurement generated by the cross product of these two primary vectors to increase estimation accuracy:

$$\mathbf{B} = \sum_{i=1}^{N} \alpha_i \mathbf{s}_{Bi} \mathbf{v}_{Ni}^T \tag{3.1}$$

Weight α_i were found as the inverse of the accuracy of the sensors exploited in section 2.4.2 and 2.4.3. For the third measurement, an arbitrary accuracy value of 0.1° was assumed:

$$\hat{\alpha}_1 = \frac{1}{0.02} = 50, \qquad \hat{\alpha}_2 = \frac{1}{0.05} = 20, \qquad \hat{\alpha}_3 = \frac{1}{0.1} = 10 \tag{3.2}$$

$$\text{and then normalized as:} \quad \alpha_i = \frac{\hat{\alpha}_i}{\hat{\alpha}_1 + \hat{\alpha}_2 + \hat{\alpha}_3} \tag{3.3}$$

The matrix $\mathbf{B}$ is used to define the auxiliary terms $\mathbf{S} = \mathbf{B} + \mathbf{B}^T$, $\mathbf{Z} = [B_{23} - B_{32}, \quad B_{31} - B_{13}, \quad B_{12} - B_{21}]^T$, and $\sigma = \mathrm{tr}[\mathbf{B}]$. These elements form the matrix $\mathbf{K}$:

$$K = \begin{bmatrix} S - \sigma I & Z \\ Z^T & \sigma \end{bmatrix} \tag{3.4}$$

The final attitude quaternion $\mathbf{q}_I$ is determined by calculating the eigenvectors of $\mathbf{K}$. The eigenvector associated with the largest eigenvalue is chosen as the solution, as it maximizes the attitude gain function.

3.2 Control Laws

3.2.1 De-tumbling

The goal of de-tumbling was to reduce spacecraft angular velocity, with respect to inertial reference frame, below a given threshold. It was the initial stage of the mission and only gyroscope was used as a sensor.
A simple binary control logic was implemented, based on [2], where the pairs of thrusters are activated to reduce maximum angular velocity component at a given time. The torque T acting on the spacecraft is define by the set of equations:

$$\mathbf{T} = -M_{i_{max}} sign(\omega_{i_{max}}) e_{i_{max}}, \tag{3.5}$$

$$i_{max} = arg(max|\omega_i|), \tag{3.6}$$

$$M = \begin{bmatrix} 2lF \sin\alpha \\ 2lF \cos\alpha \\ -2xF \sin\alpha + 2xF \cos\alpha \end{bmatrix}, \tag{3.7}$$

where $i = \{1, 2, 3\}$, F is the thrust provided by a single thruster, and $e_{i_{max}}$ forms an orthonormal basis in the body frame. The geometrical variables are defined in (Figure 2.11).
The maneuver turned out to be robust and able to decelerate the spacecraft to a sufficient velocity (see figure 4.7), which is strongly dependent on the maximum frequency of the reaction of the control system.

3.2.2 Slew Maneuver

Following the de-tumbling phase, the spacecraft must align its attitude with the LVLH reference frame in order to accomplish the mission objectives. The re-pointing phase was assumed to start from 40 seconds after satellite deployment. The transition from the initial attitude to the LVLH frame involved large-angle rotations, which could not be accurately handled by linear control techniques and therefore required a non-linear control approach. In order to do do this, the following non-linear control law was adopted:

$$\mathbf{u} = -k_1\boldsymbol{\omega}_I - k_2\mathbf{q} \tag{3.8}$$

where $\mathbf{u}$ denotes the control torque applied to the spacecraft, ω_I is the angular velocity of the satellite with respect to the inertial frame, expressed in the body frame, and $\mathbf{q}$ represents the attitude error quaternion describing the rotation of the body frame with respect to the LVLH reference frame. The gains k_1 and k_2 were selected as the average of the values derived in the linear case (see 3.12) to preserve performances in both control schemes:

$$k_1 = \frac{K_{14} + K_{25} + K_{36}}{3} = 0.3617 \qquad k_2 = \frac{K_{11} + K_{22} + K_{33}}{3} = 0.0925 \tag{3.9}$$

These values allowed for a slew maneuver with limited angular rates (under 5 $°/s$) to ensure the proper operation of the sun and horizon sensors. Once the slew maneuver was completed and the body rate $\boldsymbol{\omega}$ was sufficiently small, the nonlinear control law was deactivated, and the system was transitioned to the LQR-based nominal pointing control.

3.2.3 Nominal Pointing

Nominal pointing control was based on the linear-quadratic regulator (LQR), the control input was $\mathbf{u} = \mathbf{Kx}$. State of the system was described by $\mathbf{x} = [q_1, q_2, q_3, \omega_1, \omega_2, \omega_3]^T$, where $\mathbf{q}$ represented rotation of body frame with respect to LVLH and ω the body rate with respect to the LVLH frame represented in the body frame. Following set of linearized equations around the tracking condition $[0; 0; 0; 0; 0; 0]$ was derived, taking into consideration the gravity gradient [9]:

$$\dot{\mathbf{x}} = \mathbf{Ax} + \mathbf{Bu}, \quad \mathbf{A} = \begin{bmatrix} \mathbf{0}_{3\times3} & \mathbf{A}^{12}_{3\times3} \\ \mathbf{A}^{21}_{3\times3} & \mathbf{A}^{22}_{3\times3} \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} \mathbf{0}_{3\times3} \\ diag(\frac{1}{I_x}, \frac{1}{I_y}, \frac{1}{I_z}) \end{bmatrix}, \tag{3.10}$$

$$\mathbf{A}_{12} = diag(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}), \quad \mathbf{A}_{21} = diag(f_{41}, f_{52}, f_{63}), \quad \mathbf{A}_{22} = \begin{bmatrix} 0 & 0 & f_{46} \\ 0 & 0 & 0 \\ -f_{46} & 0 & 0 \end{bmatrix}, \tag{3.11}$$

where $f_{41} = -(8n^2(I_y - I_z))/I_x$, $f_{52} = -(6n^2(I_x - I_z))/I_y$, $f_{63} = 2n^2(I_x - I_y))/I_z$, $f_{46} = n(I_x - I_y + I_z))/I_x$. The initial values of $\mathbf{Q}$ and $\mathbf{R}$ were set using Bryson's rule, where $Q_{ii} = 1/max(error_{state})$ and $R_{ii} = 1/max(error_{input})$. However, during the tuning process, $\mathbf{R}$ was increased in order to decrease the noisy oscillating behavior of the system. The final selected values were:

$$\begin{aligned}
q_{x,\max} &= 0.037 & q_{y,\max} &= 0.037 & q_{z,\max} &= 0.0524 \\
\omega_{x,\max} &= 0.00233\,\mathrm{rad/s} & \omega_{y,\max} &= 0.00233\,\mathrm{rad/s} & \omega_{z,\max} &= 0.0104\,\mathrm{rad/s} \\
u_{x,\max} &= 0.0005\,\mathrm{Nm} & u_{y,\max} &= 0.0005\,\mathrm{Nm} & u_{z,\max} &= 0.0005\,\mathrm{Nm}
\end{aligned}$$

For small angles, the chosen limits for the quaternion components correspond to a maximum angular error of 4.24° about x and y axes, and of 6° about z axis. By applying the built-in MATLAB LQR function with the previously defined matrices, the following gain matrix $\mathbf{K}$ was obtained:

$$\mathbf{K} = \begin{bmatrix} 0.0134 & 0 & -0.0002 & 0.4398 & 0 & 0.0004 \\ 0 & 0.0134 & 0 & 0 & 0.4402 & 0 \\ 0.0002 & 0 & 0.0095 & 0.0013 & 0 & 0.2051 \end{bmatrix} \tag{3.12}$$

3.3 Management Logic

To deactivate the propulsion system after the de-tumbling phase, a termination logic based both on a time constraint and a threshold on $\boldsymbol{\omega}_I$ was implemented: due to the thrusters' effectiveness in decelerating the spacecraft, a maximum firing duration of 30 seconds was set; beyond that, a maximum allowed $\boldsymbol{\omega}_I$ to switch off the thrusters was placed at 0.0007 rad/s. The slew maneuver, governed by the non-linear control logic, was automatically activated after 40 seconds from the deployment of the satellite. Subsequently, the switch to the LQR nominal pointing mode was managed with a time-based logic. A settling time of 800 seconds was selected, as numerical analysis indicated that the residual angular velocity $\boldsymbol{\omega}$ at this stage is sufficiently small to ensure the validity of the linear control.

4. Simulation results

4.1 Baseline Simulation (Nominal Case)

A baseline simulation was performed in `Simulink` with the following initial conditions:

$$\boldsymbol{\omega}_{I,0} = [12 \quad -5 \quad 10] \quad ^\circ/s \tag{4.1}$$

$$\mathbf{q}_{I,0} = [0 \quad 0 \quad 0 \quad 1] \tag{4.2}$$

The simulation was conducted using the solver `ode8` with a fixed step size of 0.1 s. To enhance the physical meaning of the results, the attitude error relative to the LVLH frame is presented in terms of Euler angles rather than the quaternion error $\mathbf{q}$.

4.1.1 Uncontrolled Case

The first section of the results analyzes the uncontrolled time evolution of the angular velocity error $\boldsymbol{\omega}$ and the Euler angles over a single orbital period. External perturbations, specifically gravity gradient and magnetic field torques, were included in the disturbance model.

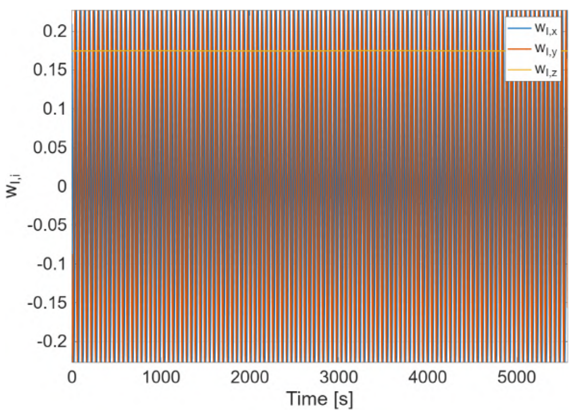


Figure 4.1: Angular rates of the S/C with respect to LVLH with no control

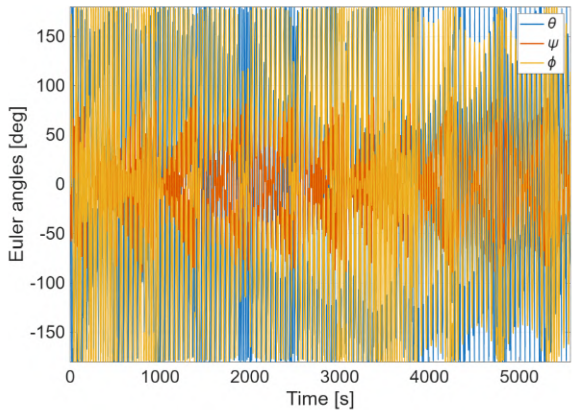


Figure 4.2: Euler angles components with no control

The figures above clearly show that the spacecraft shows a chaotic motion if no control is active.

4.1.2 Controlled case

De-tumbling and Slew Maneuver

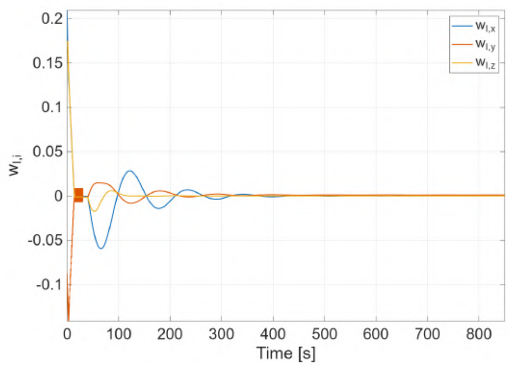


Figure 4.3: Angular rates of the S/C during de-tumbling and slew

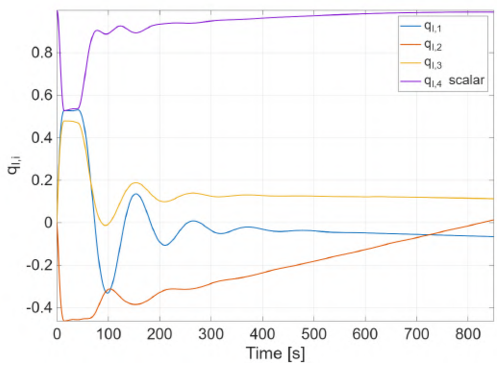


Figure 4.4: Quaternion's components during de-tumbling and slew

The simulation results demonstrated a rapid de-tumbling phase (lasting less than 30 sec) due to the high efficiency of the thruster in handling high initial angular rates $\boldsymbol{\omega}_{I,0}$. This resulted in a decay of $\boldsymbol{\omega}_{\boldsymbol{I}}$ to near-zero values, implying that the satellite was stabilized with a fixed orientation in the inertial frame. This brief activation period was sufficient to bring the spacecraft's total angular momentum within the non-saturated operating envelope of the Control Moment Gyroscopes. Moreover prolonged thruster activation beyond this threshold was avoided to prevent unnecessary fuel consumption. Following the initiation of the slew maneuver initial variations in $\boldsymbol{\omega}_I$ and $\mathbf{q}_I$ were observed, arising from the fact that the spacecraft was stationary in a fixed inertial orientation. To achieve alignment with the LVLH reference frame, large angle rotation were required.

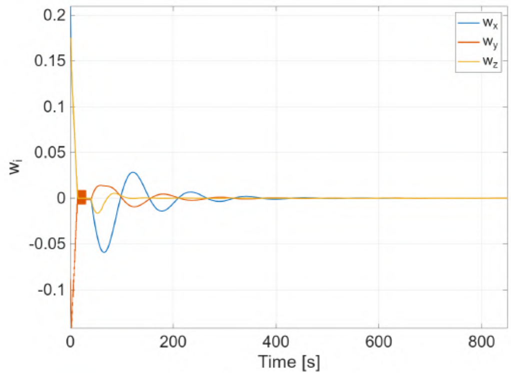


Figure 4.5: Angular rates of the S/C with respect to LVLH during de-tumbling and slew

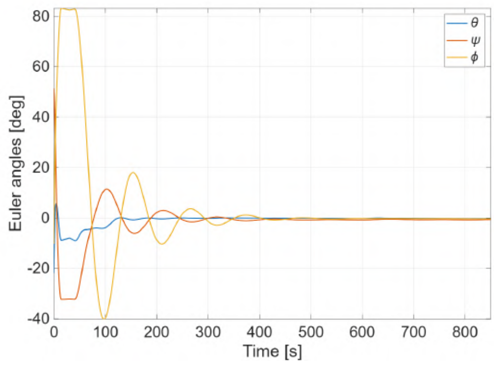


Figure 4.6: Euler angles during de-tumbling and slew

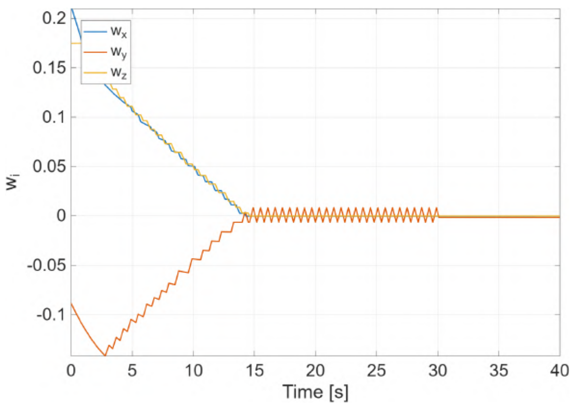


Figure 4.7: Detail on $\boldsymbol{\omega}$ during de-tumbling phase

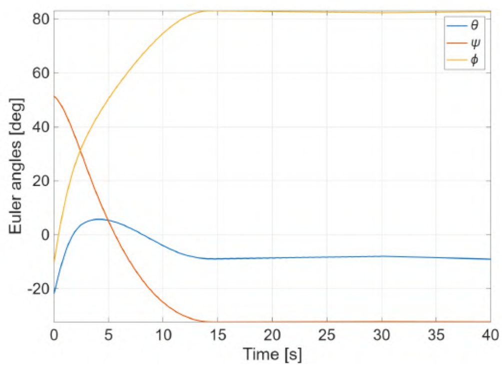


Figure 4.8: Detail on Euler angles during de-tumbling

During the de-tumbling phase, significant magnitudes variations of $\boldsymbol{\omega}$ and Euler angles were observed, as the thrusters were engaged to stabilize the satellite with respect to the inertial frame rather than the target LVLH frame. During the brief interval in which the satellite was inertially stabilized, the Euler angles, as the quaternion error's components, exhibited nearly-constant values. This behavior was attributed to the slow evolution of the LVLH reference frame relative to the inertial frame during the holding period preceding the slew maneuver. With the previously mentioned (3.9) values of k_1 and k_2 the satellite steered with an oscillating behavior toward the target position. The application of the non-linear control law ensured that both the angular rate error $\boldsymbol{\omega}$ and the quaternion error $\mathbf{q}$ converged to zero in approximately 0.143 orbits (800 sec).

Tracking maneuver

This phase constituted the majority of the simulated period. The primary objective was to maintain precise nadir pointing utilizing the control torque provided by the CMG cluster.

Throughout this stage, the actuator carried out only minor corrective adjustments to neutralize environmental disturbances.

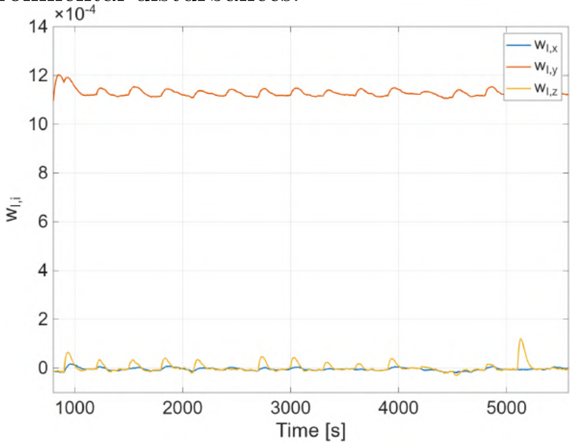


Figure 4.9: Angular rates of the S/C with respect to inertial frame during tracking phase

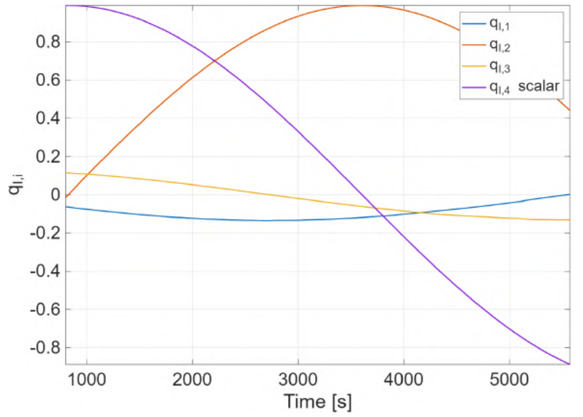


Figure 4.10: Quaternion's components during tracking phase

The plots exhibit periodic peaks every 300 seconds, which are attributable to the reset of the gyroscope measurement performed by the attitude determination algorithm using sun and horizon sensor data. Furthermore, in figure 4.9, the y-component of inertial angular velocity $\boldsymbol{\omega}_I$ settles at a constant value of approximately 0.0011 rad/s. This offset corresponds to the orbital mean motion $n \simeq 0.0011$ rad/s, confirming that the spacecraft is correctly tracking the rotating LVLH frame. Note that the LVLH frame rotates about the y-axis.

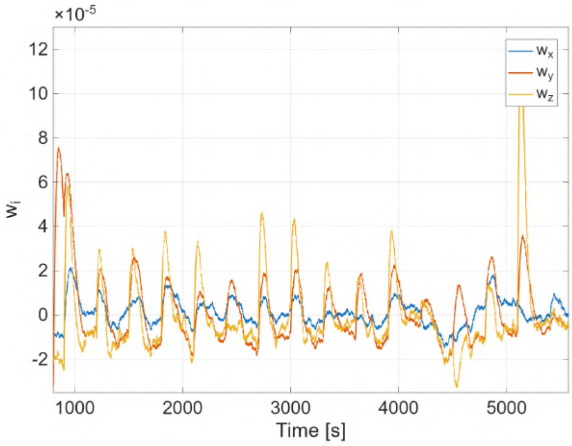


Figure 4.11: Angular rates of the S/C with respect to LVLH frame during tracking phase

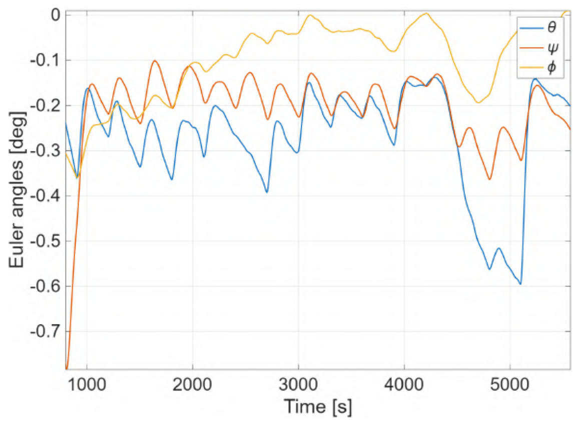


Figure 4.12: Euler angles during tracking phase

The figures show that $|\boldsymbol{\omega}|$ (angular velocity error magnitude) consistently remained under 1.2×10^{-4} rad/s, while Euler angles stayed under 0.8°. These values provide evidence that the control laws perform effectively and satisfy the pointing accuracy requirements established for the mission in 3.2.3.

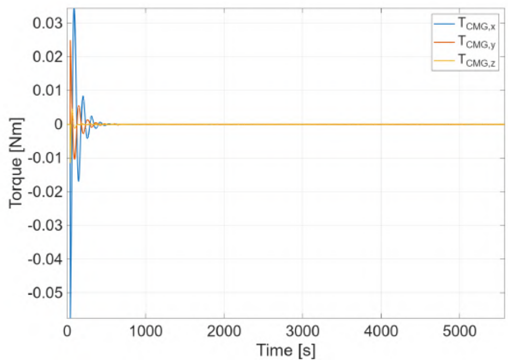


Figure 4.13: Control torque's components provided by the CMG cluster

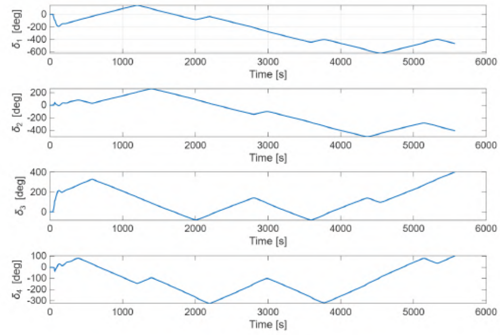


Figure 4.14: δ angles of the gimbal axes over the simulated period

The temporal evolution of CMG gimbal δ angles was characterized by slow variations, consistent with the low angular rates maintained both during the slew maneuver and the tracking phase. Furthermore, the cluster demonstrated the capability to traverse singular conditions without locking, thanks to the null motion implementation. The torque provided by the actuator remained well below the saturation limit of 1.1 Nm, with a peak value of only 0.06 Nm. This significant safety margin confirms the adequate sizing of the CMG cluster, ensuring sufficient control authority to handle unexpected disturbances or larger slew rates. The control effort was primarily concentrated in the initial simulation phase to execute the slew maneuver and align the spacecraft with the LVLH frame.

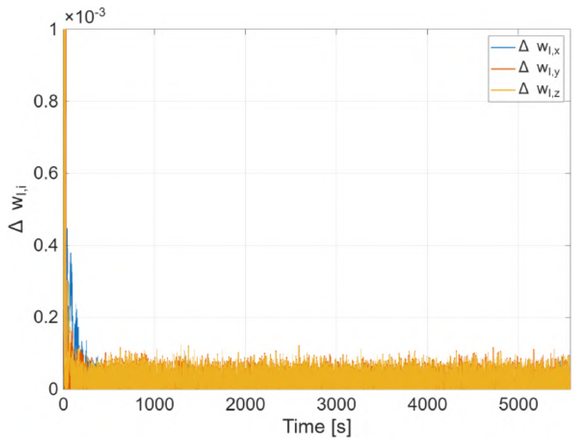


Figure 4.15: Difference between real and measured $\boldsymbol{\omega}_I$

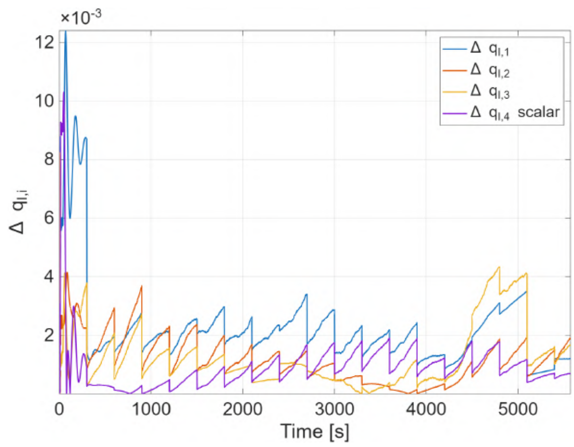


Figure 4.16: Difference between real and measured $\mathbf{q}_I$

Excluding the initial part of the simulation, the errors between the real quantities and the measured ones remained constantly small, dominated primarily by sensor noise and bias instability.

4.2 Robustness Analysis (Monte Carlo)

To validate the global stability of the control architecture, a Monte Carlo campaign consisting of $N = 50$ simulations, each spanning one full orbital period, was performed. The analysis focused on the system's ability to recover from randomized initial conditions, specifically varying the initial attitude quaternion (uniformly distributed) and the initial angular velocity vector (random direction with magnitude up to 15 deg/s). Plant parameters (e.g., inertia tensor, mass properties, and actuator saturation limits) and sensor noise characteristics were kept at their nominal values to isolate the convergence performance.

Performance Overview The Fig. 4.22 aggregates the results. The top row displays the pointing error evolution on the three body axes relative to the mission requirements (red dashed lines). Despite the high initial tumbling rates, the controller consistently stabilizes the satellite within approximately 500 s in all instances, with no divergence observed. Post-convergence, the error trajectories on all axes remain strictly bounded within the defined limits (4.24° for x_{body} and y_{body}, and 6.00° for z_{body}), demonstrating consistent compliance with pointing requirements during the pointing phase.

Statistical Validation The robustness is quantified in the bottom row of Fig. 4.22. The angular velocity error (bottom-left, log scale) demonstrates convergence to a residual noise floor of $10^{-3} \div 10^{-4}$ deg/s. The histogram (bottom-right) specifically analyzes the worst-case pointing error across all runs. The maximum observed error is $\approx 0.38°$, which provides a safety margin of over $10\times$ against the tightest requirement (4.24°). This confirms that the control law is robust regardless of the deployment conditions.

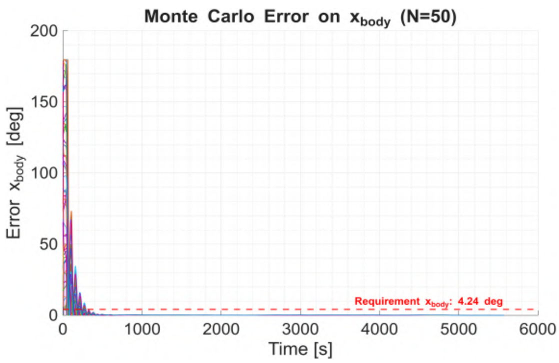


Figure 4.17: Error x_{body}

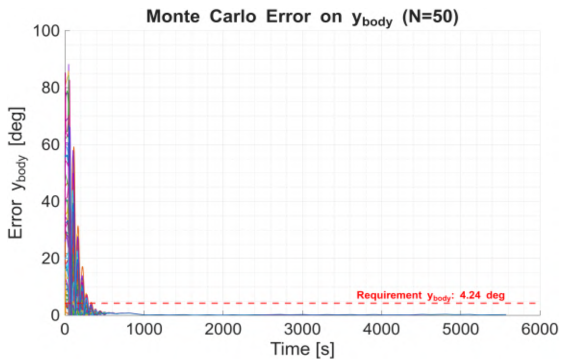


Figure 4.18: Error y_{body}

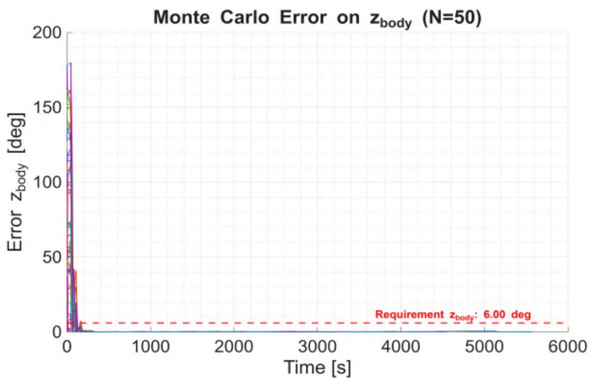


Figure 4.19: Error z_{body}

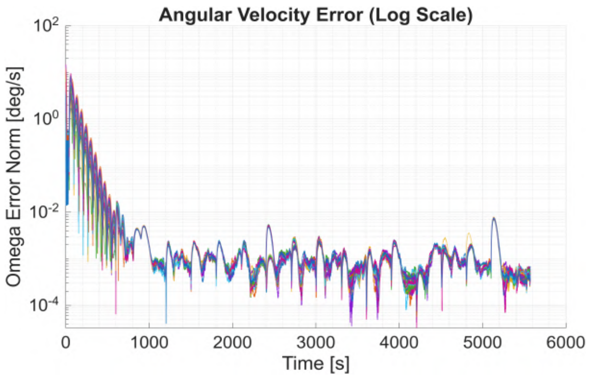


Figure 4.20: Ang. Velocity Error (Log)

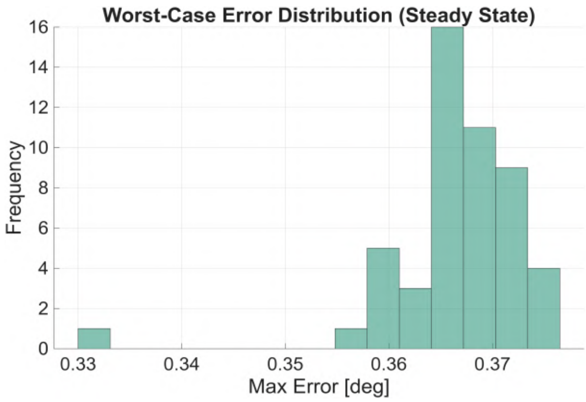


Figure 4.21: Worst-case Error Distr.

Figure 4.22: Monte Carlo Analysis results ($N = 50$). The system converges from random initial tumbling to errors well below the red dashed requirements.

5. Conclusions

5.1 Verification of requirements

The objective of this project was to simulate the complete attitude dynamics and control loop of a microsatellite. The model demonstrated excellent performance, ensuring that attitude errors were kept under the defined thresholds with a significant margin. The results were validated through Monte Carlo analysis, numerical simulations and comparison with established physical known model. However, simplified assumptions were adopted; consequently, the model can be improved.

5.2 Future Developments

The developed model is consistent with the physical environment and represents an appropriate preliminary phase of the spacecraft design. However, the simulation could be refined to achieve higher fidelity. For instance, additional environmental disturbances could be included: the most relevant one for a LEO orbit satellite is air drag, but also J2 zonal harmonic, solar radiation pressure play a significant role. Moreover, sensors modeling could be to account non-nominal conditions. For example, including the field of view constraints for the horizon and sun sensor would yield a more accurate representation of the attitude determination system. Similarly, the fidelity of the actuator models could be increased by introducing dissipative effects. Specifically, accounting for pressure drops and finite valve response times in the thruster lines and mechanical friction and vibrations in the CMG would capture more realistic variations in control performance. An alternative approach to singularities management would involve the use of a Variable Speed CMG (VSCMG) cluster. Unlike the standard constant-speed architecture, this systems introduces four additional degrees of freedom by allowing independent modulation of each wheel's angular velocity. Consequently, when approaching a singularity, the VSCMG acts as a reaction wheel to avoid it, while in nominal conditions it operates as a conventional CMG cluster [3]. Finally, in a more advanced design stage, a Hardware-in-the-Loop testing could be implemented to validate the numerical simulation.

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1. Interplanetary Explorer Mission

1.1 Mission overview

1.1.1 Mission requirements and objectives

The objective of this study is the preliminary analysis of an interplanetary mission departing from Venus, destined to reach asteroid N.147080, exploiting an intermediate Gravity Assist (GA) on Earth to reduce the overall cost in terms of Δv. The imposed temporal requirements define an extremely wide launch window, spanning from January 1, 2030, to January 1, 2060 (30 years).

The primary optimization goal is to minimize the total mission cost Δv_{tot}, defined as the sum of the impulsive velocity changes required for departure, the powered flyby maneuver, and the final orbital insertion: $\Delta v_{tot} = \Delta v_{dep} + \Delta v_{GA} + \Delta v_{arr}$.

1.1.2 Assumptions and Physical Models

The analysis is conducted under the following simplifying assumptions, consistent with a preliminary feasibility study:

- **Patched Conics Method**: Trajectories are modeled as heliocentric Keplerian arcs (patched conics method), assuming that the planetary Spheres of Influence (SOI) are point-like.
- **Impulsive maneuvers**: All velocity changes are considered instantaneous.

1.2 Design Strategy: The Space Pruning Approach

1.2.1 The Curse of Dimensionality

Optimizing an interplanetary trajectory with an intermediate flyby is a problem characterized by three main degrees of freedom (3-DoF): the departure date (t_{dep}), the flyby date (t_{ga}), and the arrival date (t_{arr}). Considering the breadth of the assigned time window (30 years), an exhaustive search approach (Brute Force Grid Search) with a daily temporal resolution would require the evaluation of over 10^9 possible combinations. Such computational cost makes a blind search unfeasible. To address this complexity, a Space Pruning strategy was adopted, based on the physics of the problem and inspired by global optimization methodologies developed by the ESA Advanced Concepts Team [2].

1.2.2 Objective Function Bounding

Before initiating the search algorithm for trajectories with flybys, it was necessary to establish a rigorous criterion to identify and discard non-competitive solutions. According to the Branch and Bound approach described by Addis et al. (2011) [1], search efficiency depends on defining an Upper Bound (UB) for the cost function. The direct Venus-Asteroid mission was chosen as the "Incumbent Solution". As highlighted by Labunsky et al. (1998) [3], the inclusion of an intermediate Gravity Assist is justified only if it lowers the total cost below that of the direct Baseline Mission.

- **Baseline Computation**: A preliminary Lambert search on the direct transfer within the 2030 - (2030 + $3T_{syn}^{VA}$) (See sec. 1.2.3) interval identified a minimum reference cost $\Delta v_{direct} \approx 14.8$ km/s.
- **Pruning Condition**: Consequently, the constraint $UB = 14.8$ km/s was imposed. Following the concept of Admissible Region introduced by Vasile and De Pascale (2006) [4], the algorithm was designed to immediately prune any computation branch where the partial cost (even of just the first leg) exceeds this value.

1.2.3 Search Space Pruning

Having defined the energy limit, the reduction of the temporal domain was achieved by exploiting the periodicity of planetary configurations, as suggested by Izzo (2010) [2].

The Venus → Earth leg drives the mission scheduling. Since both orbits are nearly circular and coplanar, optimal geometries repeat cyclically according to the Synodic Period defined as $1/T_{syn} = |1/T_{Venus} - 1/T_{Earth}|$.

Traditionally, an "Anchor Window" strategy involves identifying the first optimal opportunity and propagating it mathematically by integer multiples of T_{syn}. In this study, a generalized variation of this strategy is applied. To account for the orbital eccentricity—which causes slight variations in the exact timing of the windows—and to ensure no feasible opportunities are missed due to theoretical approximations, the algorithm performs a coarse feasibility scan directly over the entire 30-year period, effectively "discovering" the real windows rather than just predicting them.

1.2.4 Feasibility Filter and Window Identification

The results of the coarse feasibility scan are visualized via a scattered porkchop plot (See Figure 1.1), which effectively subdivides the search space into discrete "islands" of opportunity where the cost is lower than the Upper Bound (UB).

A filtering algorithm is then applied to these islands. The algorithm detects the boundaries of the valid data blocks, converting the continuous temporal domain into a set of discontinuous, bounded grids defined by specific minimum and maximum values ($[t_{dep_{min}}, \ t_{dep_{max}}]$ and $[ToF_{min}, \ ToF_{max}]$) for each identified window.

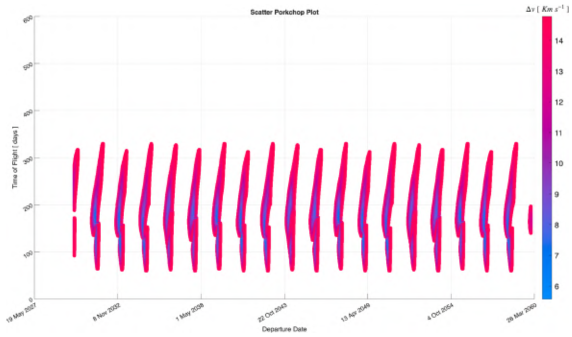


Figure 1.1: Porkchop plot of Venus-Earth leg

1.2.5 Pruned Grid Search

Once the search space has been pruned into specific windows, the grids for the optimization routine are generated.

- **Leg 1 (Venus → Earth)**: Dense grids are generated strictly within the bounds of each identified window.
- **Leg 2 (Earth → Asteroid)**: The grid is defined using the Earth-Asteroid synodic period as a maximum transfer time value.

A nested triple loop then evaluates the total mission cost for every combination within these refined grids. This step identifies the single global minimum candidate among all windows, providing the necessary initial guess for the subsequent local refinement.

1.3 Refined Optimization

1.3.1 Local Grid Search Setup

Following the Pruned Grid Search, the global minimum candidate is identified. However, due to the discretization of the grid, this solution represents a "valley" rather than the precise mathematical minimum. To refine this result and provide an accurate initial guess (x_0) for the gradient-based optimizer, a **Detailed Grid Search** is performed. This involves a high-resolution scan within a narrow temporal window (±20 days) centered around the best candidate found in the pruning phase.

1.3.2 Optimization Solvers

To converge to the precise local minimum, the solution is passed to the `MATLAB` function `fmincon.m`. Unlike grid search methods, which are bound to discrete steps, this solver operates in a continuous domain using a gradient-based approach. The algorithm is initialized with the best result from the Detailed Grid Search and minimizes the objective function while strictly satisfying the imposed non-linear constraints. Specifically, the required precision:

Algorithm	Tol_{Step}	$\text{Tol}_{\text{Constraint}}$	$\text{Tol}_{\text{Function}}$	$\text{Max}_{\text{Func Eval}}$
`SQP`	1×10^{-14}	1×10^{-9}	1×10^{-14}	10000

1.3.3 Mission Objective Function

The objective is to minimize the total mission cost (Δv_{tot}) with respect to the optimization vector $\mathbf{x} = [t_{dep}, \ \Delta t_1, \ \Delta t_2]$, representing the departure date and the transfer times for the two legs, where t_{dep} is the departure date (MJD2000), Δt_1 is the Time of Flight for the Venus-Earth leg, and Δt_2 is the Time of Flight for the Earth-Asteroid leg.

1.3.4 Mission Constraint Function

To ensure flight safety and avoid atmospheric drag, a non-linear constraint enforces a minimum pericenter radius $r_p \geq R_E + h_{safe}$, where $R_E = 6371$ km is Earth's radius and $h_{safe} = 300$ km is the altitude margin.

1.4 Mission Analysis Results

1.4.1 Algorithm Setup and Grid Definitions

The optimization algorithm was executed with a total run time of approximately 4 hours. The search space discretization strategies adopted for the different phases of the algorithm are detailed below.

Direct transfer Venus → Asteroid (147080) The baseline for the direct transfer was computed over a departure window spanning from $\mathrm{Dep_{min}}$ to $\mathrm{Dep_{min}} + 3\mathrm{T^{VA}_{syn}}$. The Time of Flight (ToF) domain was bounded between 100 days and $2\mathrm{T^{VA}_{syn}}$.

First Leg Venus → Earth The preliminary feasibility scan for the first leg covered the entire 30-year mission window ($\mathrm{Dep_{min}}$ to $\mathrm{Arr_{max}}$). The flight time was constrained between 60 days and one Venus-Earth synodic period ($\mathrm{T^{VE}_{syn}}$).

Brute Force Search A coarse global search validated the pruning logic over the following domain: Departure $\in [\mathrm{Dep_{min}},\ \mathrm{Arr_{max}} - \mathrm{T^{VE}_{syn}}]$, Leg 1 ToF $\in [60,\ \mathrm{T^{VE}_{syn}}]$ days, and Leg 2 ToF $\in [60,\ \mathrm{T^{EA}_{syn}}]$ days.

Pruned Grid Search (PGS) In this phase, the departure dates and Leg 1 flight times were restricted to the specific "islands" identified by the pruning algorithm (see Sec. 1.2.3). Consequently, the grid bounds vary dynamically for each identified window. The second leg (Earth → Asteroid) was explored with flight times ranging from 60 days to $\mathrm{T^{EA}_{syn}}$.

Refined Grid Search The final high-resolution grid search focused on a narrow local neighborhood around the best candidate found in the PGS phase, defined as follows:

Departure	Leg 1 ToF	Leg 2 ToF
$PGS_{Dep} \pm 20$ days	$PGS_{TOF_1} \pm 20$ days	$PGS_{TOF_2} \pm 20$ days

Reference Constants: The search grid relies on the mission window $[\mathrm{Dep_{min}},\ \mathrm{Arr_{max}}]$ = [01/01/2030, 01/01/2060] and the synodic periods $T^{VE}_{syn} = 583.92$ d, $T^{EA}_{syn} = 505.06$ d, and $T^{VA}_{syn} = 270.82$ d.

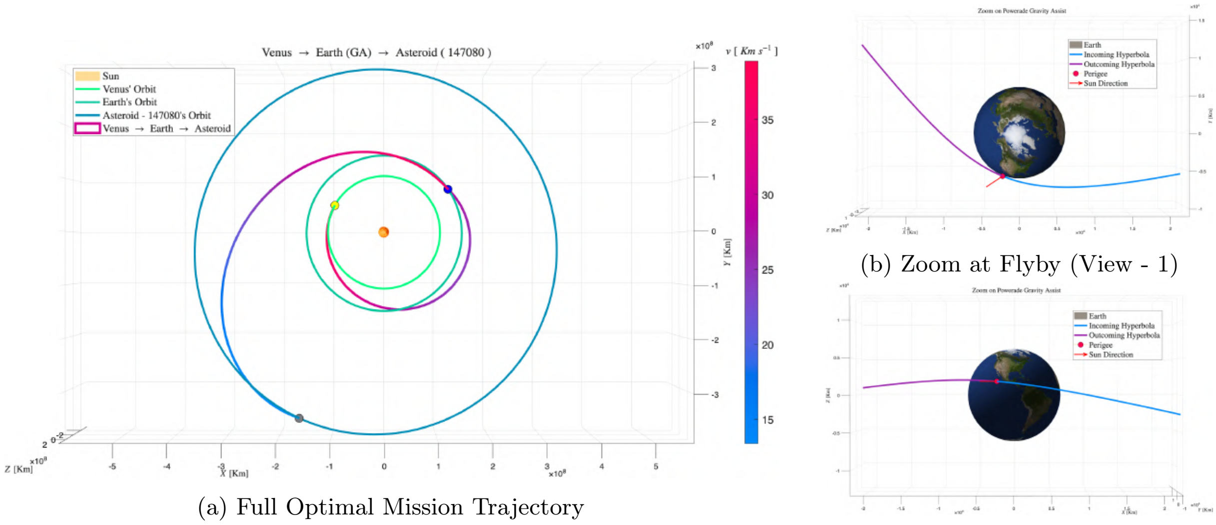
1.4.2 Optimal Trajectory Data

The optimization process successfully converged to a specific mission window. The key epochs and duration of the optimal solution are reported below:

Departure	Earth Flyby	Arrival	Duration	Δt_{fb} (SOI)
27-02-2057 | 23:49:59 (MJD: 20877.49)	26-10-2057 | 15:50:25 (MJD: 21118.16)	13-03-2059 | 06:41:07 (MJD: 21620.78)	**743d 6h 51m 7s**	**73.451h**

1.4.3 Transfer Arcs Characterization

The optimal trajectory resulting from the `fmincon` optimization is illustrated in Figure 1.2. The mission profile consists of two heliocentric arcs connected by a powered gravity assist at Earth.



(a) Full Optimal Mission Trajectory

(b) Zoom at Flyby (View - 1)

(c) Zoom at Flyby (View - 2)

Figure 1.2: Full Heliocentric Trajectory | Zoom at Flyby

1.4.4 Mission Cost Breakdown

The total optimized cost (Δv_{tot}) and its specific components are detailed below:

Maneuver	**Cost [km/s]**
Departure (Venus) (Δv_{dep})	4.04417
Powered Gravity Assist (Earth) (Δv_{GA})	0.52234
Arrival (Asteroid) (Δv_{arr})	4.22099
Total Mission Cost ($\boldsymbol{\Delta v_{tot}}$)	**8.78750**

1.5 Flyby Analysis

The critical phase of the mission is the Gravity Assist (GA) performed at Earth. This maneuver is modeled using the "Patched Conics" approximation, assuming the spacecraft traverses a hyperbolic trajectory relative to Earth within its Sphere of Influence (SOI).

1.5.1 Geometry of the Hyperbola

The geometry of the flyby is entirely determined by the heliocentric velocity vectors at the discontinuity point between the two legs:

- **Incoming Hyperbolic Excess Velocity ($\vec{v}_\infty^-$):** Derived from the arrival velocity of the first leg relative to Earth ($\vec{v}_{arr,Leg1} - \vec{v}_{Earth}$).
- **Outgoing Hyperbolic Excess Velocity ($\vec{v}_\infty^+$):** Derived from the departure velocity of the second leg relative to Earth ($\vec{v}_{dep,Leg2} - \vec{v}_{Earth}$).

The required turning angle δ, representing the deflection needed for the plane change, is computed via the dot product as $\delta = \arccos\left((\vec{v}_\infty^- \cdot \vec{v}_\infty^+)/(||\vec{v}_\infty^- \cdot \vec{v}_\infty^+||)\right)$.

1.5.2 Powered vs. Unpowered Gravity Assist

In a purely ballistic (unpowered) flyby, the magnitude of the relative velocity is conserved ($||\vec{v}_\infty^-|| = ||\vec{v}_\infty^+||$), and the planet only rotates the velocity vector. However, for this mission, a **Powered Gravity Assist** model was adopted to maximize flexibility. This implies that a propulsive impulse (Δv_{GA}) is applied at the pericenter of the hyperbola to change the energy of the orbit. The cost of this maneuver is calculated as the difference between the required velocity at pericenter to exit on the second leg (v_p^+) and the natural velocity reached at pericenter from the first leg (v_p^-), expressed as $\Delta v_{GA} = |v_p^+ - v_p^-| = \left| \sqrt{||\vec{v}_\infty^+||^2 + 2\mu_E/r_p} - \sqrt{||\vec{v}_\infty^-||^2 + 2\mu_E/r_p} \right|$.

The solution gives a ratio of 0.0676 (6.76%) between the pericenter impulse and the total flyby Δv. This result confirms the maneuver's high efficiency.

Moreover, since the maneuver is executed in proximity of the Line of Nodes of the asteroid's orbit, the cost of this phase is minimized.

1.5.3 Operational Constraints Verification

Post-optimization analysis confirms that the trajectory respects the safety limit against atmospheric drag, with a pericenter altitude $h_p = r_p - R_E \geq 300$ km, ensuring physical feasibility and avoiding thermal damage.

1.6 Conclusions

This study successfully demonstrated the feasibility and efficiency of a Venus-to-Asteroid mission exploiting an intermediate Earth Gravity Assist. The primary objective of minimizing the total mission cost (Δv_{tot}) was achieved through a rigorous "Space Pruning" strategy, which effectively addressed the "Curse of Dimensionality" inherent in a 30-year search window.

The analysis yielded the following key conclusions:

- **Effectiveness of Space Pruning:** The hierarchical reduction of the search space—from the global direct-transfer baseline to synodic window identification and finally to pruned grid searches—allowed for the efficient evaluation of millions of potential trajectories. This approach successfully isolated the global optimum without necessitating a prohibitive brute-force computation.

- **Benefit of the Gravity Assist:** The comparison between the direct transfer baseline ($\approx$ 14.8 km/s) and the optimized flyby trajectory ($\Delta v_{tot} \approx 8.8$ km/s) confirms the significant energetic advantage of the Earth flyby. The inclusion of the intermediate leg reduced the total impulse requirement by approximately **6 Km/s**.

- **Optimization Refinement:** The transition from discrete grid searches to the continuous gradient-based solver (`fmincon`) proved crucial. The local optimizer successfully refined the solution, pinpointing the precise epoch times and reducing the residual cost to the mathematical local minimum.

- **Operational Feasibility:** The final trajectory satisfies all imposed physical constraints. Specifically, the powered gravity assist at Earth respects the minimum altitude safety margin ($h_p \geq 300$ km), ensuring that the maneuver is not only theoretically efficient but also operationally viable in terms of atmospheric drag avoidance. Specifically, the final and optimal altitude h_p is slightly above 300 Km.

In conclusion, the identified trajectory represents a robust preliminary mission profile, offering a high-efficiency alternative to direct transfers and validating the use of pruning algorithms for complex interplanetary mission design.

2. Planetary Explorer Mission

2.1 Introduction

2.1.1 Mission Overview

The PoliMi Space Agency has proposed a Planetary Explorer Mission aimed at performing detailed observations of celestial bodies within the Solar System. As part of the mission analysis team, the objective of this study is to carry out a comprehensive orbit analysis and ground track estimation for the spacecraft. The project requires characterizing the nominal orbit, evaluating the effects of gravitational perturbations, comparing different numerical propagation methods, and designing an orbit modification to achieve a repeating ground track for targeted observations.

2.1.2 Selected Scenario

Among the proposed mission profiles, this report focuses on the Departure Planet Observation scenario. Before the injection into the interplanetary transfer trajectory, the spacecraft is tasked with an observation campaign of the departure planet. For this specific mission analysis, the central body is Venus. This choice implies that the gravitational environment is dominated by Venus' gravitational parameter (μ_V) and its specific zonal harmonic coefficients (mainly J_2), while the primary third-body perturbation source is the Sun.

2.2 Nominal Orbit Analysis

2.2.1 Initial Keplerian Elements

The analysis begins with the propagation of the nominal orbit starting from a defined set of Keplerian elements. The motion is described in a Venus-Centred Inertial (VCI) Equatorial frame, where the fundamental plane coincides with Venus' equator and the principal direction points towards the vernal equinox equivalent for Venus at the reference epoch.

The initial epoch for the simulation is set to March 18, 2053, at 07:27:51. The initial orbital parameters assigned for the mission are:

a [km]	**e** [-]	**i** [deg]	**Ω** [deg]	**ω** [deg]	$\boldsymbol{\theta_0}$ [deg]
181 712	0.2083	133	143	60	0

It is important to note that the initial true anomaly is set to $\theta_0 = 0°$, placing the spacecraft at the pericenter of its orbit at the start of the simulation.

2.2.2 Nominal Orbit Visualization

The nominal orbit, propagated assuming a Keplerian two-body problem (unperturbed motion), is visualized in Figure 2.1. Given the semi-major axis of approximately 1.8×10^5 km, the spacecraft operates in a high-altitude orbit. The orbital period T, calculated via the relation $T = 2\pi\sqrt{a^3/\mu_V}$, corresponds to approximately 9.883 days. This long period suggests that the spacecraft will spend a significant portion of its time at high altitudes, potentially exposing it to third-body perturbations from the Sun more intensely than a Low Venus Orbit (LVO).

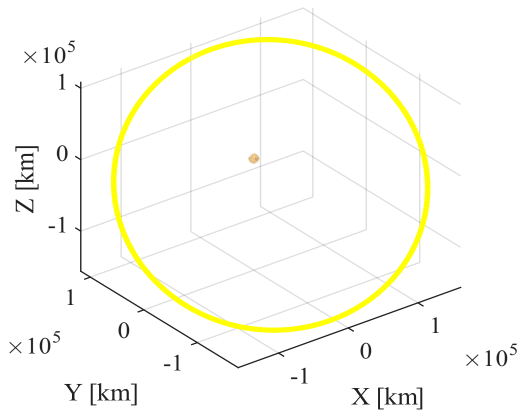


Figure 2.1: Nominal Venus-centred orbit representation in the VCI frame.

2.3 Perturbation Modeling

To accurately predict the long-term evolution of the spacecraft's orbit, the relevant perturbing accelerations have been modeled and included in the equations of motion. Specifically, the non-spherical mass distribution of Venus and the gravitational influence of the Sun have been considered.

2.3.1 Venus Zonal Harmonics (J_2)

The oblateness of Venus is modeled by considering the second zonal harmonic coefficient, J_2. The corresponding perturbing acceleration $\boldsymbol{a}_{J2}$ is expressed in the Cartesian frame. Since zonal harmonics depend only on latitude (and distance) and are independent of longitude, the formula can be applied directly using the position vector $\boldsymbol{r} = x\hat{\boldsymbol{i}} + y\hat{\boldsymbol{j}} + z\hat{\boldsymbol{k}}$ in the inertial frame, provided the z-axis is aligned with the planet's rotation axis.The acceleration components are given by:

$$\boldsymbol{a}_{J2} = \frac{3}{2}\frac{J_2 \mu_V R_V^2}{r^5}\left[\left(5\frac{z^2}{r^2} - 1\right)x\hat{\boldsymbol{i}} + \left(5\frac{z^2}{r^2} - 1\right)y\hat{\boldsymbol{j}} + \left(5\frac{z^2}{r^2} - 3\right)z\hat{\boldsymbol{k}}\right]$$

where: μ_V is Venus' gravitational parameter, R_V is Venus' mean equatorial radius, $r = \sqrt{x^2 + y^2 + z^2}$ is the distance from the planet center,

2.3.2 Sun Third-Body Perturbation

The gravitational attraction exerted by the Sun is modeled as a third-body perturbation. The acceleration acting on the spacecraft relative to Venus, $\boldsymbol{a}_{3B}$, is calculated as the difference between the acceleration induced by the Sun on the spacecraft and the acceleration induced by the Sun on Venus:

$$\boldsymbol{a}_{3B} = \mu_{Sun}\left(\frac{\boldsymbol{r}_{Sun} - \boldsymbol{r}}{|\boldsymbol{r}_{Sun} - \boldsymbol{r}|^3} - \frac{\boldsymbol{r}_{Sun}}{|\boldsymbol{r}_{Sun}|^3}\right)$$

where: μ_{Sun} is the Sun's gravitational parameter, $\boldsymbol{r}_{Sun}$ is the position vector of the Sun with respect to Venus, $\boldsymbol{r}$ is the position vector of the spacecraft with respect to Venus.

2.4 Ground Track Analysis

This section characterizes the ground track coverage and evaluates the feasibility of a repeating orbit. The analysis is performed first under ideal Keplerian conditions and subsequently including J_2 and Solar perturbations. The observation windows were selected as:

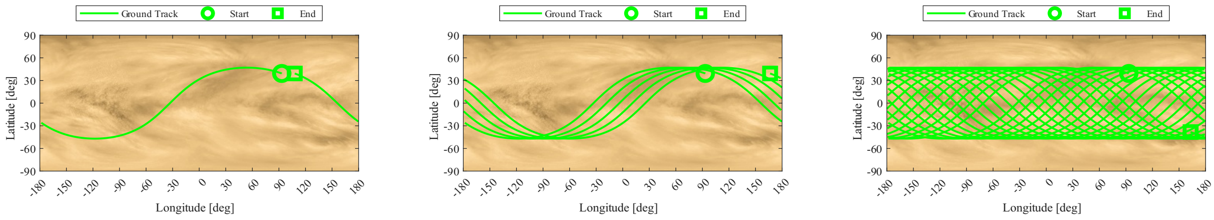
1. Single Orbit ($1T$): To visualize the fundamental closed shape of the track.
2. Multiple Orbits ($5T$): To highlight the longitudinal spacing between consecutive passes.
3. Full Venus Rotation (243 days): To assess the global coverage achievable in one Venusian sidereal day.

For the repeating orbit, a ratio $k : m = 34 : 1$ was selected, yielding a semi-major axis $a_{rep} \approx 146409$ km ($T_{rep} \approx 7.15$ days).

2.4.1 Unperturbed Scenario

- Nominal Orbit ($a = 181712$ km): Figure 2.2 shows the natural Eastward drift of $\approx 14.6°$ per orbit caused by Venus' retrograde rotation. The track covers the planet densely over 243 days but does not repeat.

- Repeating Orbit ($a_{rep} = 146409$ km): The designed orbit behaves perfectly. As illustrated in Figure 2.3, the track closes exactly upon itself after one Venusian sidereal day, confirming the validity of the $k = 34$ design.



(a) Single orbital period (b) Five orbital periods (c) Full Venusian sidereal day

Figure 2.2: Nominal Ground Track (Unperturbed Scenario)

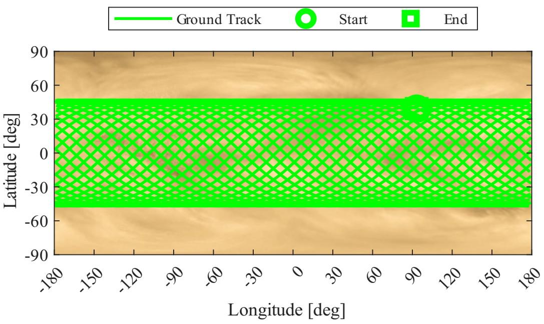
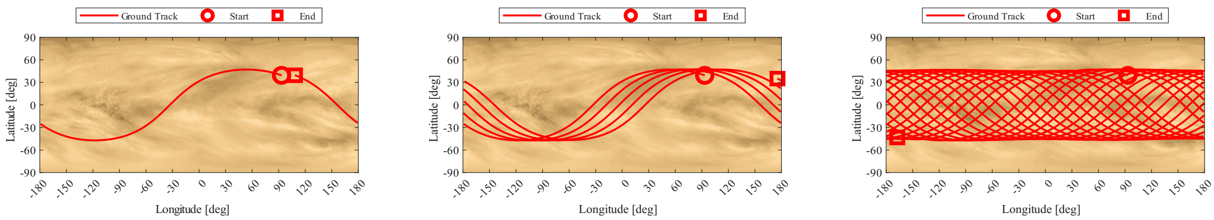


Figure 2.3: Repeating Ground Track (Unperturbed Scenario)

2.4.2 Perturbed Scenario

When the assigned perturbations (J_2 + Sun) are included, the orbital elements evolve over time, breaking the symmetry of the ground track.

- Nominal Orbit: The perturbations induce secular and long-period variations that modify the nodal period. While the global coverage pattern remains qualitatively similar to the unperturbed case, the exact crossing points shift over time (see Figure 2.4).
- Repeating Orbit Analysis: The strict repetition condition does not hold under perturbations. As shown in Figure 2.5, the track fails to close after 243 days. This occurs because the repeating condition relies on a constant nodal period. However, J_2 induces a secular precession of the RAAN ($\dot{\Omega}$) and perigee ($\dot{\omega}$), while the Solar third-body force adds periodic variations. These effects alter the "effective" period of the satellite relative to the rotating planet, causing a cumulative longitudinal drift ($\Delta\lambda_{drift}$) that prevents the track from overlapping with the initial pass.



(a) Single orbital period (b) Five orbital periods (c) Full Venusian sidereal day

Figure 2.4: Nominal Ground Track (Perturbed Scenario)

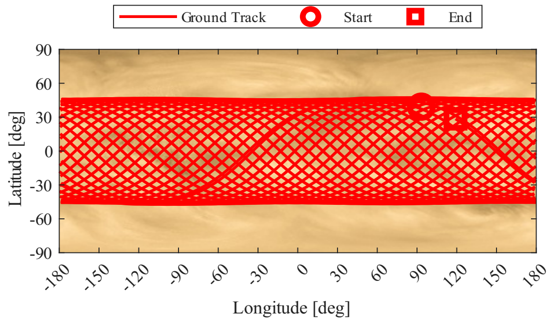


Figure 2.5: Repeating Ground Track (Perturbed Scenario)

2.5 Elements Propagation and Behaviour

Up to this section, the trajectory analysis (e.g., Ground Tracks) was performed using Cartesian propagation, which is ideal for visualizing the spacecraft's position in space. However, to analyze the long-term evolution of the orbit and isolate the effect of perturbations, it is necessary to introduce the Gauss Planetary Equations.

2.5.1 Numerical Methods Comparison

- **Cartesian Propagation (Cowell's Method)**: This method involves the direct numerical integration of the equations of motion in the inertial frame ($\ddot{\boldsymbol{r}} = -\frac{\mu}{r^3}\boldsymbol{r} + \boldsymbol{a}_{pert}$). While conceptually straightforward, it requires converting the state vector back to Keplerian elements at each time step to analyze the orbital geometry.

- **Gauss Planetary Equations**: This approach integrates the rates of change of the orbital elements ($\dot{a}$, $\dot{e}$, $\dot{i}$, Ω, ω, θ) directly. This method is introduced, since Gauss equations provide immediate physical insight into how specific forces affect the orbit shape and orientation. Furthermore, calculating the variations of elements directly is more suitable for the subsequent long-term analysis, where mean elements will be extracted to study secular trends.

2.5.2 Validation

Before proceeding with the long-term analysis using Gauss equations, a validation step is performed to ensure consistency with the established Cartesian baseline. The absolute difference $|\alpha_{Gauss} - \alpha_{Cart}|$ was computed for all six orbital elements over one Venusian year.
Figure 2.6 illustrates the results. The error remains extremely low and bounded for all parameters:

- **Semi-major axis** (a): The discrepancy remains bounded below 10^{-6} km (millimeter scale). Since a is directly linked to the orbital energy, this negligible error confirms that both propagators maintain energy consistency.

- **Eccentricity and Angles**: Eccentricity and angles exhibit errors in the order of 10^{-10} to 10^{-14}, attributable to machine precision. This confirms that the Gauss implementation is numerically equivalent to the Cartesian one, validating its use for the Long-Term Evolution analysis in the next section.

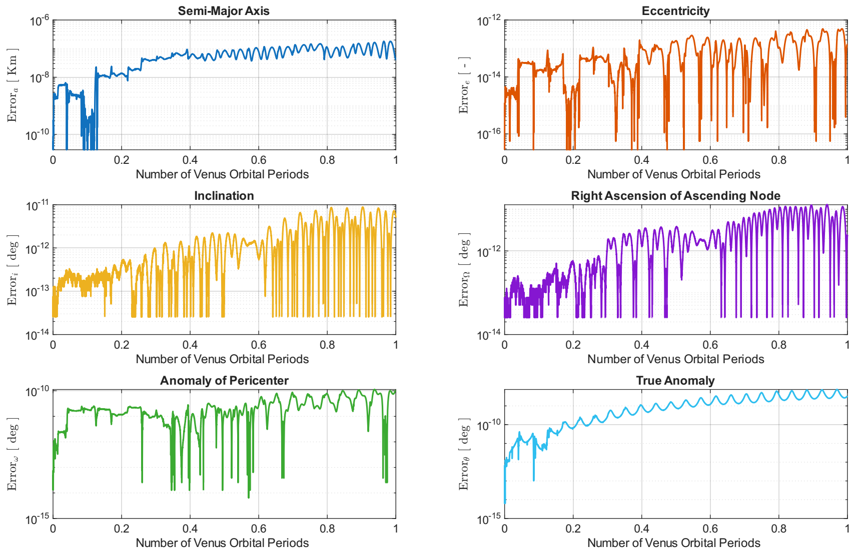


Figure 2.6: Errors | Cartesian vs. Gaussian

2.5.3 Orbital Elements Evolution Analysis

The history of the Keplerian elements was analyzed to isolate the effects of the different perturbations. The simulation covers a period of one Venusian year ($T_{Venus} \approx 225$ days), sufficient to observe both secular trends and long-period solar effects.

Figure 2.7 displays the evolution of the orbital elements, comparing the effect of J_2 only, Sun Third-Body only and the Total perturbation.

- **Semi-major axis** (a): The total variation remains stable around the nominal value, confirming that no orbital decay occurs (conservative forces only). J_2 induces only short-period oscillations, while the Sun adds a minor periodic variation.
- **Eccentricity** (e) **and Inclination** (i): No secular trend is observed. The evolution is dominated by the Solar perturbation, which drives long-period oscillations (period $\approx 1/2T_{Venus}$), visible in the overlap between the "3B" and "Total" curves.
- **RAAN** (Ω) **and Perigee** (ω): The evolution is dominated by the J_2 secular drift. Ω exhibits a constant nodal regression, while ω precesses linearly. The Solar perturbation adds only a small periodic modulation on top of this trend.
- **True Anomaly** (θ): Being the fast variable, it cycles $0 \to 360°$ rapidly. The perturbations induce minor phase shifts relative to the unperturbed prediction.

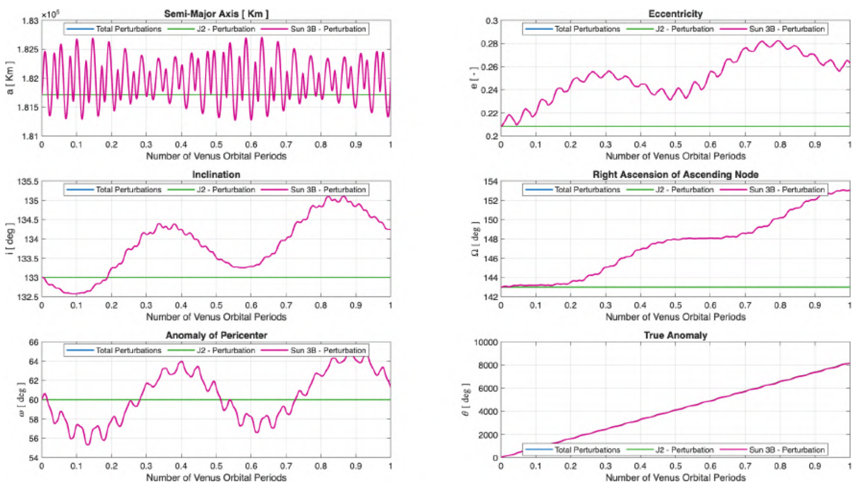


Figure 2.7: All Perturbations | J2 vs. Sun-3B vs. Total

2.6 Orbit Evolution

Gravitational perturbations induce secular variations in the orbital elements. To appreciate the magnitude of these effects and the resulting shift in the trajectory (e.g., nodal precession), the orbit is propagated and visualized over 100 periods:

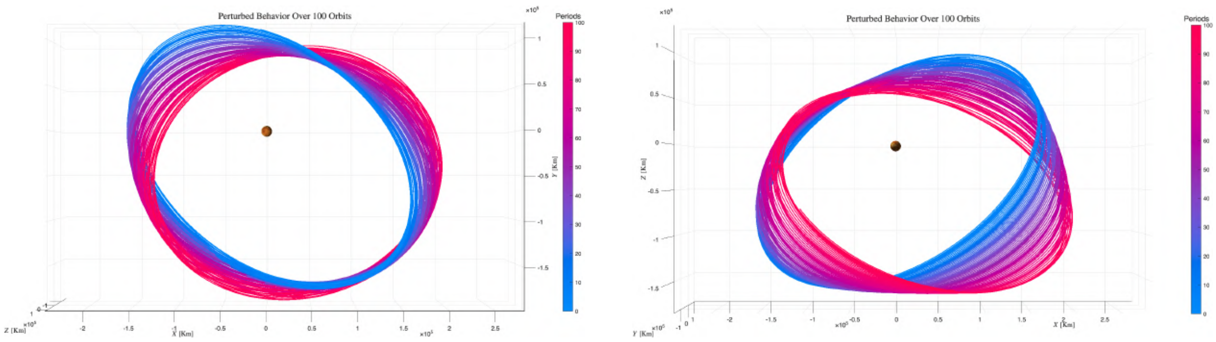


Figure 2.8: Perturbed Orbit | 100 Periods

2.7 High Frequencies Filtering

Orbital perturbations affect the Keplerian elements on different time scales. The time history of any element can be viewed as the superposition of three distinct components:

- **Short-period variations**: High-frequency oscillations occurring within a single orbital period (T), driven primarily by the zonal harmonics of the central body (e.g., J_2).
- **Long-period variations**: Oscillations with periods significantly longer than T, typically related to the rotation of the line of apsides or the motion of disturbing third bodies.
- **Secular Variations**: Monotonic trends that persist over very long time spans.

To isolate the long-term and secular behavior required for the mission analysis, it is necessary to filter out the high-frequency components. This is achieved by applying a low-pass filter to the time series of the orbital elements. In this analysis, a moving average filter (implemented via the `MATLAB` function `movmean`) is employed. The critical design parameter for this filter is the window size (T_w), which determines the cut-off frequency.

2.7.1 Selection Of Filtering Windows

The choice of the window size is physically motivated by the periodicity of the perturbations to be suppressed.

- **Suppression of J_2 Effects (Short-Period)**: The J_2 perturbation depends on the satellite's position relative to the equator and repeats cyclically with the satellite's true anomaly. Consequently, the fundamental frequency of these variations corresponds exactly to the satellite's orbital period (T_{sat}). To remove these short-period oscillations and reveal the underlying evolution, the filtering window is set to match the nominal orbital period of the spacecraft: $T_{w,J2} = T_{sat}$. This specific window width is chosen because short-period effects average out over exactly one complete revolution (as the true anomaly sweeps from 0 to 2π). Therefore, a window of a single period is the most efficient choice to smooth the signal without masking the physical long-term dynamics.
- **Suppression of Third-Body Effects (Long-Period)**: Perturbations due to a third body (the Sun) vary with the relative geometry between the planet (Venus) and the disturbing body. This geometry repeats with the period of Venus's revolution around the Sun ($T_{VenusYear}$).To filter out these long-period oscillations and isolate the pure "Secular"

trend, a second, wider window is selected corresponding to the period of the perturbing body: $T_{w,3rdBody} \approx T_{VenusYear}$. Applying this filter removes the annual oscillations caused by the Sun, leaving only the secular drift of the orbital elements.

2.7.2 Comparison Between Filtered and Un-Filtered Elements

Figure 2.9 illustrates the effectiveness of the filtering process.

- **Semi-Major Axis (a)**: The unfiltered signal (Blue) shows high-frequency noise. However, the filtered evolution (Pink) is almost perfectly constant. This aligns with theory, as J_2 does not cause secular variations in the semi-major axis, confirming the filter has correctly removed the periodic noise.
- **Eccentricity (e) and Inclination (i)**: The filtering removes the "thickness" of the line (short-period oscillations), clearly revealing the large-scale sinusoidal exchange of angular momentum driven by the third-body perturbation (Sun).
- **RAAN (Ω) and Pericenter (ω)**: The filtered lines show clear secular drifts (linear trends) combined with long-period modulations, which would be difficult to quantify in the noisy unfiltered data.
- **True Anomaly (θ)**: As the "fast variable" of the system, its evolution is dominated by the mean motion. The linear trend remains unchanged after filtering, simply reflecting the continuous revolution of the satellite around the central body.

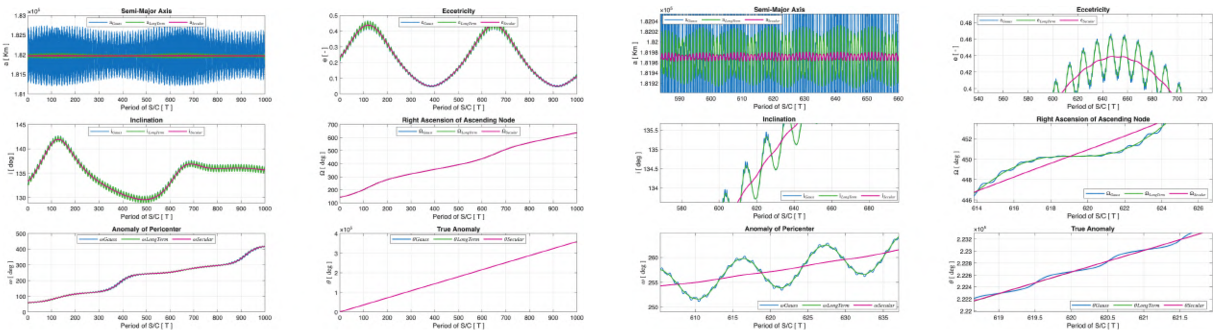


Figure 2.9: Filtering | Filtered vs. Unfiltered

2.8 Real Data Analysis

To validate the perturbation analysis in a real-world scenario, it is convenient to shift focus to objects orbiting in Earth's gravitational field.

2.8.1 LEO Object Analysis | SL-3 R/B

To analyze the perturbation environment in the Low Earth Orbit (LEO) regime, a spent rocket body has been selected. Rocket bodies are ideal candidates for this analysis due to their typically high Area-to-Mass ratio (A/m), which amplifies the effects of non-conservative forces.

Name	NORAD ID	Type	Date Range	Step
`SL-3 R/B`	13819	Rocket Body	1983-02-17 → 1989-10-09	3h

The object follows a nearly circular orbit with a mean perigee radius $r_p \simeq 6961$ km ($h_p \approx 590$ km) and a mean apogee radius $r_a \simeq 7053$ km ($h_a \approx 682$ km). The eccentricity is low ($e \simeq 0.0066$).

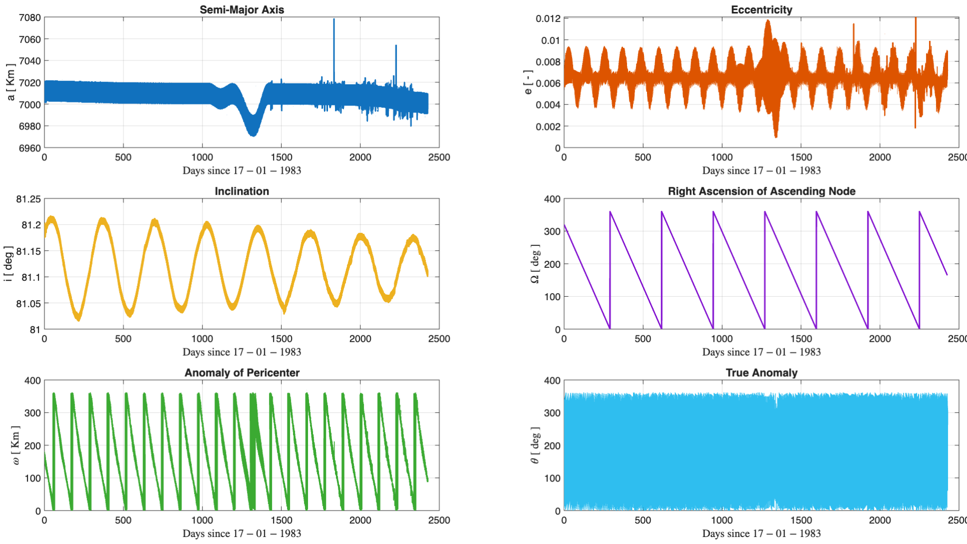


Figure 2.10: Evolution of Keplerian Elements for SL-3 R/B (LEO)

Analysis of Observed Perturbations: The evolution confirms the theoretical predictions for LEO:

1. **Atmospheric Drag:** Secular decay of semi-major axis a. Variations in slope correlate with the 6-year solar cycle activity.
2. J_2 **Effect:** Secular trends in angular elements:
   - **Nodal Regression ($\dot{\Omega} < 0$):** Westward shift of the Line of Nodes ($i < 90°$).
   - **Apsidal Precession ($\dot{\omega} \neq 0$):** Rotation of the argument of perigee over time.

2.8.2 MEO Object Analysis | Titan 3C Transtage R/B

For the second case study, an object in a high-altitude orbit (Sub-Synchronous / Super-MEO) has been chosen to observe the dynamics where drag is negligible.

Name	NORAD ID	Type	Date Range	Step
`Titan 3C Transtage R/B`	11147	Rocket Body	1984-01-01 → 1991-01-01	3h

The orbit is characterized by mean radii $r_p \simeq 36513$ km and $r_a \simeq 38843$ km. Although the eccentricity is low ($e \simeq 0.026$), the altitude ($\sim 30,000$ km) places this object well above the GPS constellation.

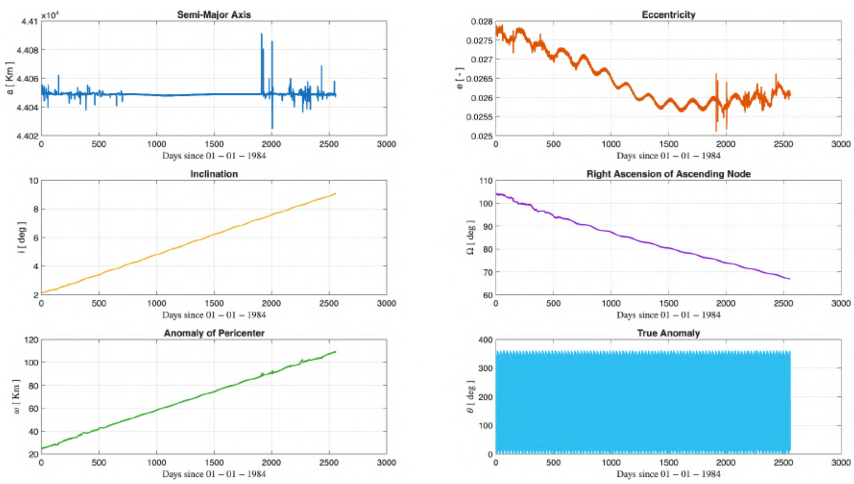


Figure 2.11: Evolution of Keplerian Elements for Titan 3C Transtage R/B (MEO)

Analysis of Observed Perturbations: Hierarchy of perturbations differs from LEO:

1. **Absence of Drag:** a remains constant (density is effectively zero).
2. **Third-Body (Dominant):** Sun and Moon drive dynamics as J_2 decays ($r^{-3.5}$).
3. **Lunisolar Cycles:** Long-period oscillations observed in e and i, unlike the stable inclination in LEO.

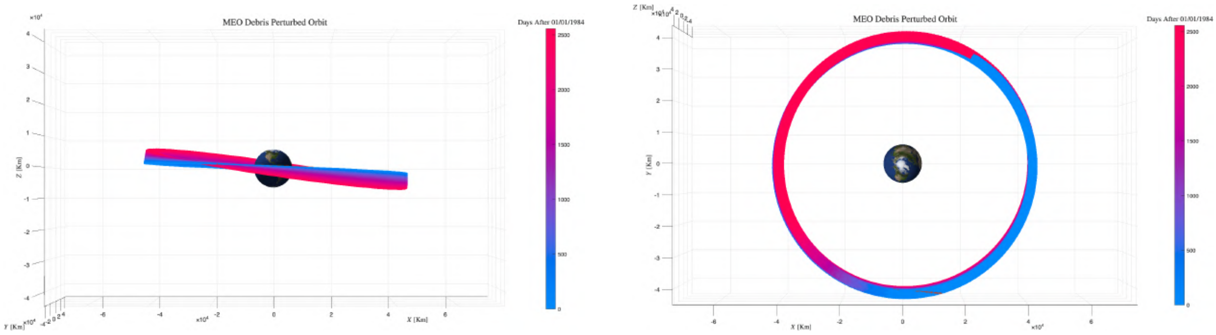


Figure 2.12: MEO Orbit

2.8.3 Conclusion: Third-Body Analysis

The gravitational influence of the Moon and Sun varies by regime. **In High-Altitude (Titan 3C Transtage R/B)**, third-body effects are significant, causing visible long-period oscillations in e and i and contributing to ω drift. **In LEO (SL-3 R/B)**, Earth's gravity dominates overwhelmingly. No long-term third-body effects are visible; dynamics are driven purely by J_2 and Atmospheric Drag.

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Solar Array Deployement

Final Report for the collaborative project of the Multibody System Dynamics course

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1. Problem statement

The aim of the work is to reproduce and study the behavior of the deployment mechanism of the solar panels on the Lunar Reconnaissance Orbiter. The system of interest consists of three panels initially found in closed configuration, attached to the body of the satellite as depicted in Figure 1.

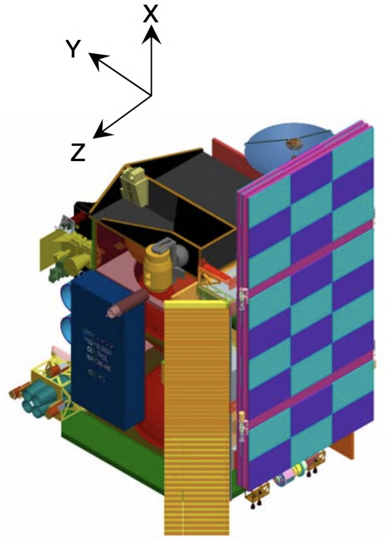

Figure 1: Closed configuration

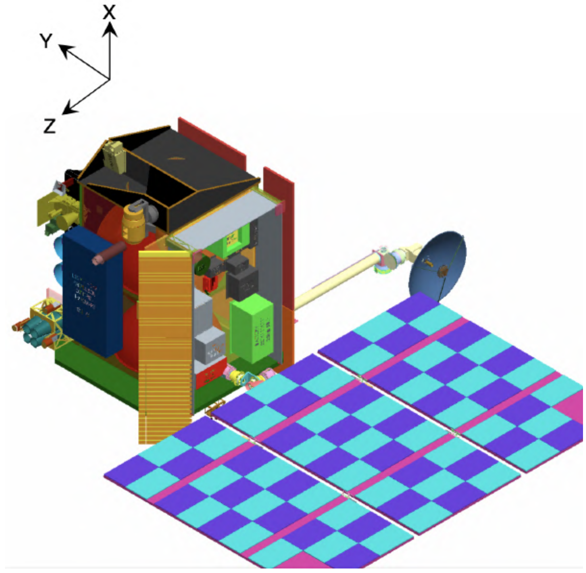

Figure 2: Open configuration

The driving mechanism consists of a DC motor, which allows the rotation of the solar array assembly, and a system of four precharged springs which, together with two dampers, regulate the side panels' opening. The sequence is shown in the following Figure 3 and Figure 4. During the first act, the closed assembly reaches the 90° of aperture in 300 seconds with respect to the body of the satellite. Ended the first motion, the left panel (seen from the satellite) starts opening thanks to the activation of the prestrained joints, through which it is linked to the central panel. Once it reaches 60° in the angular motion, the right one starts moving. The system deployment is stopped when both the side panels rotated of 180° with respect to their original position.

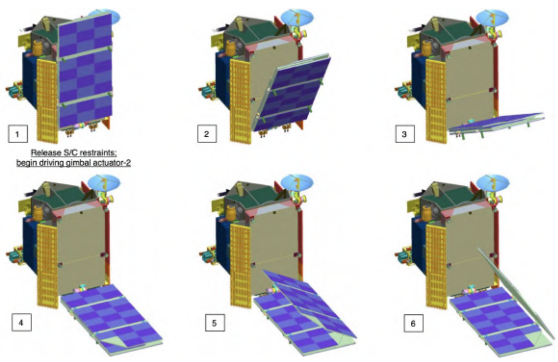

Figure 3: Deployment sequence

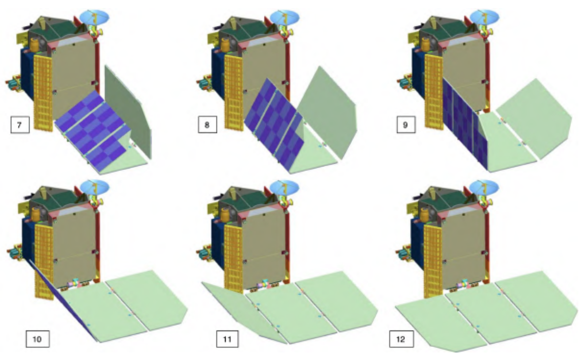

Figure 4: Deployment sequence

Three different models were therefore developed: a kinematic one on Adams, used to retrieve values of the needed couples to correctly open the panels and to size the spring-damper system; a dynamic model on MBDyn which used the found data, to retrieve positions and velocities of the points of interests and a Matlab one implementing a simplified version of the system to individuate discrepancies with the other two models and to develop the control law of the motor.

2. ADAMS Model

2.1. CAD model

In order to examine the deployment sequence in Adams, first the CAD model was needed. Based on the data provided in the LRO preliminary design review [1] panels and hinges were modelled in Siemens NX. The goal was to have a realistic model that would accurately represent the satellite. Hinges' dimensions were not provided in the document so their properties were chosen based on other geometrical values. The final model is presented below.

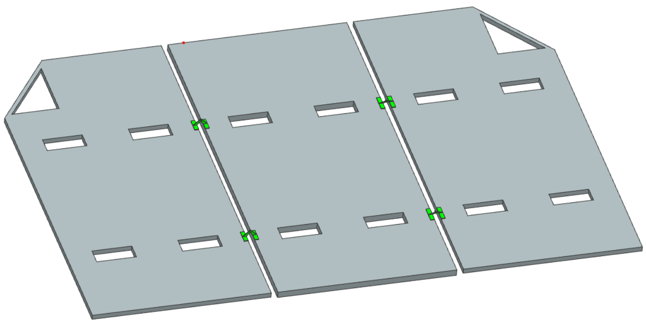

Figure 5: Panels open

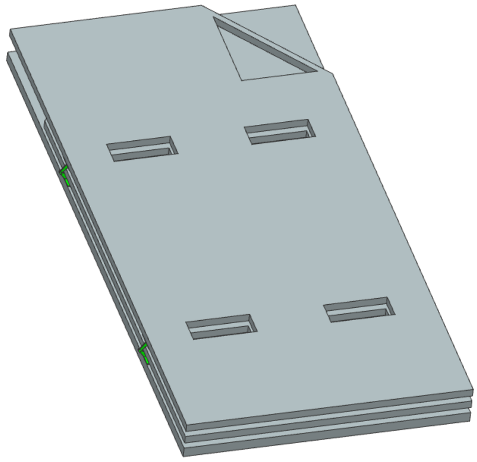

Figure 6: Panels closed

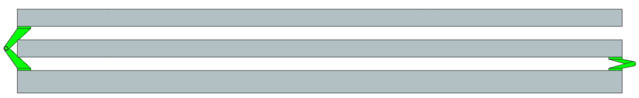

Figure 7: View from the bottom

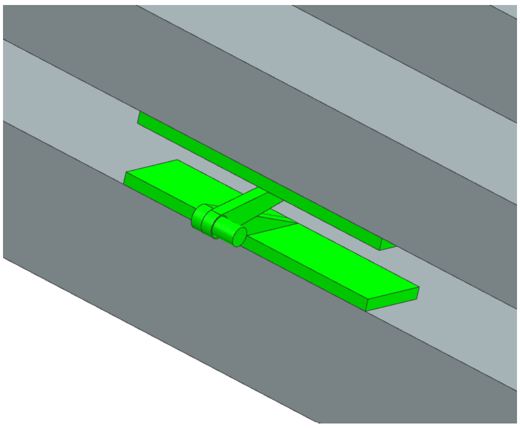

Figure 8: Hinge closeup

2.2. Basic Adams setting

Next the model in a *STEP* format was imported to Adams. Based on the volume properties of elements and masses of each panel provided in [1] the average densities were inputted.
Later on, the joints were made to connect all the elements. Firstly, the middle panel was set

as ground. Next, hinges were connected to panels by fixed joints. Later, hinges were connected by spherical joints. This decision was taken because in the end the result was identical to using double cylindrical joints, however turned out to be easier to implement it in Adams. The resultant multibody system had two degrees of freedom.
To one hinge in each pair, a spring and a damper were added. After in depth considerations and analysing the results of the simulations, the stiffness and damping coefficients were sized in order to guarantee convergence in less then 100 seconds while maintaining an angular velocity lower than 10 deg/s. Results of the sizing are shown in Table 3.
The holding mechanism was modelled as a contact force acting between panels (left panel-right panel; right panel- middle panel). For the assembly to open, a step function was designed to set the contact forces from holding to 0N in 0.1N. Each panel maximum opening angle was set to 180° and made by introducing another contact force (left panel-middle panel; right panel-middle panel).

2.3. Further Adams model development

To model the servo controlling the solar array opening sequence the following changes were made:

1. The schematic body of a satellite was designed in Adams and set as a new ground.
2. New joint mechanism between middle panel and satellite body were designed. These consisted of two independent cylindrical joints allowing the movement of the panels as it happens in reality.

The opening sequence of the solar array was implemented with the use of motion function acting on one of the servo joints. The chosen solution was a step5 function, since it guarantees very low maximum torque, described below:

$$STEP5(x, x_0, h_0, x_1, h_1)$$

x- independent variable - time
x_0- initial starting point- time= 0s
x_1- ending point- time= 300s
h_0- starting value- angle= 0°
h_1- final value- angle= 90°

The end motion results in the solar array opening from 0° to 90° in 300 s. The motion function presents a continuous first and second derivatives.
At the end the additional motion of the fully deployed panel array was added to simulate the system adjustment its position to the sun. The procedure was the same as described above.
The full opening and operating procedure simulated is described below:

1. Time= 0s- Beginning of the opening of the array
2. Time= 300s- Array reaches 90°
3. Time= 300s- The left panel holding force (locking mechanism) goes to 0N in 0.1s
4. Left panel reaches 60°
5. The right panel holding force (locking mechanism) goes to 0N in 0.1s
6. First left then right panel reach 180−° opening angle
   (Here the further animation part of simulation is described)
7. Time= 700s Panel array open further by 45° using step5 function
8. Second servo joint turns the panel around animating the sun tracking procedure

Figure 9 shows the motion imposed to the motor joint; Figure 10 shows the angular position and velocity obtained with the sized spring-damper systems.

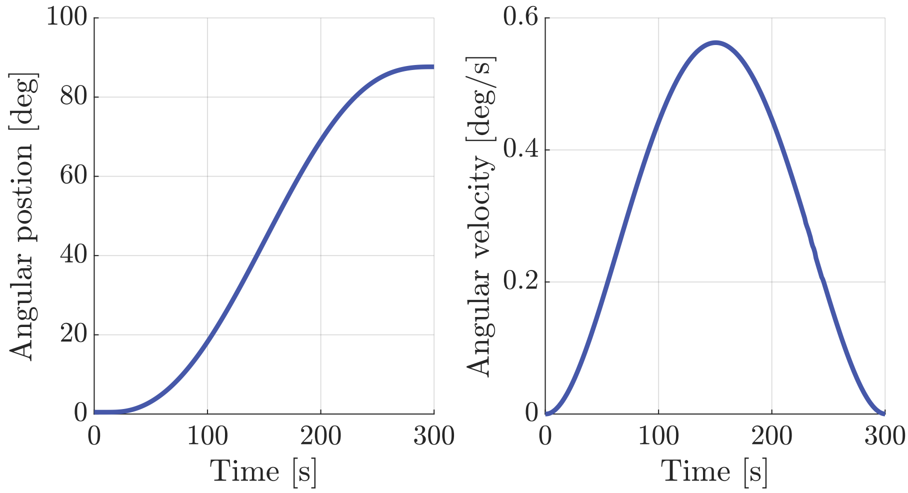

Figure 9: First Deployment

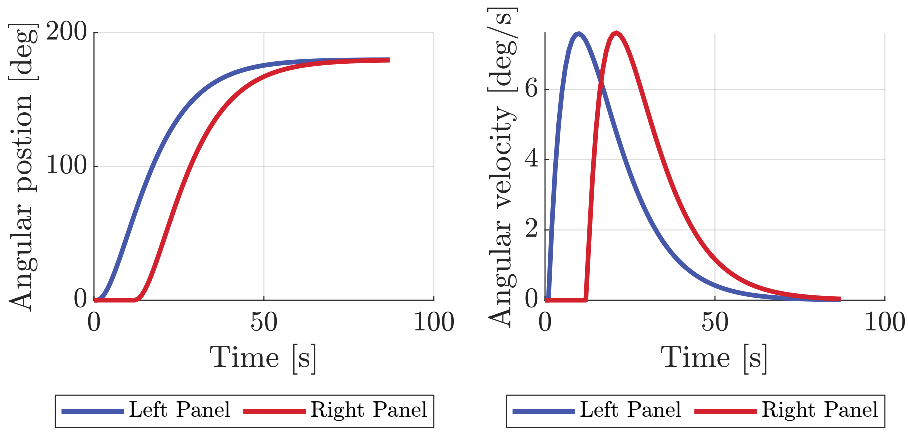

Figure 10: Second Deployment

Through the above explained and shown motion imposition it was possible to retrieve the intensity of the torque needed to correctly execute the motion. The moment curve is displayed in Figure 11.

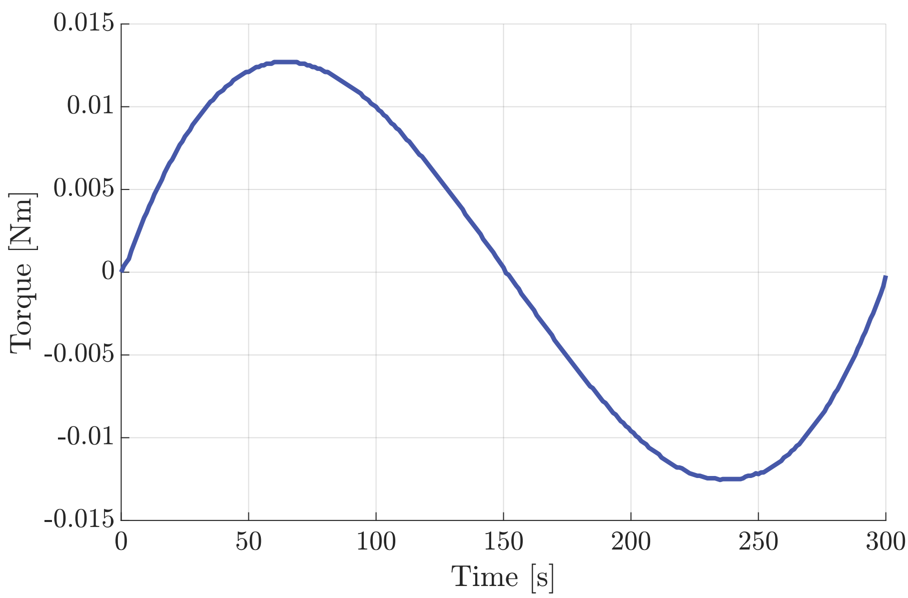

Figure 11: Torque

2.4. Simulation animation

The images below show some pictures from the animation sequence fully available at the following link.

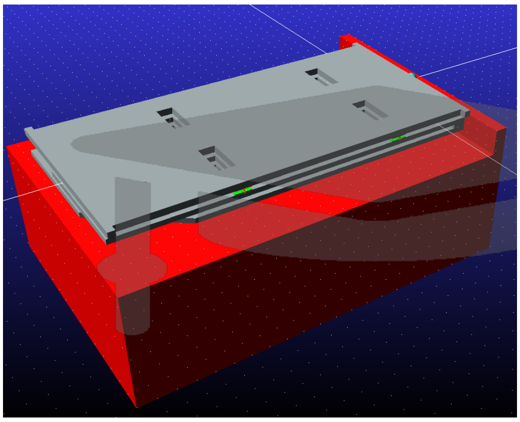

Figure 12: Closed satellite

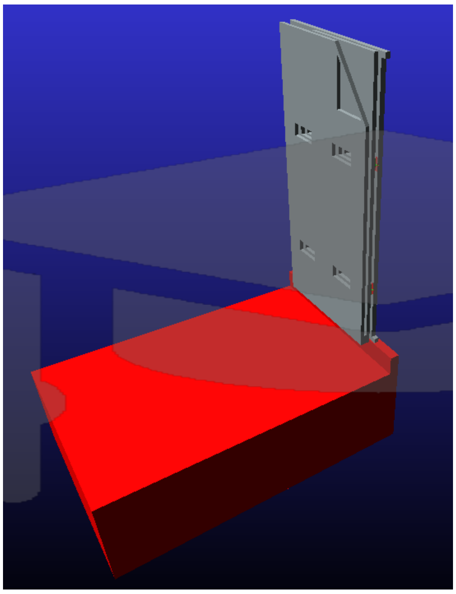

Figure 13: Sollar assemlby rotated by 90°

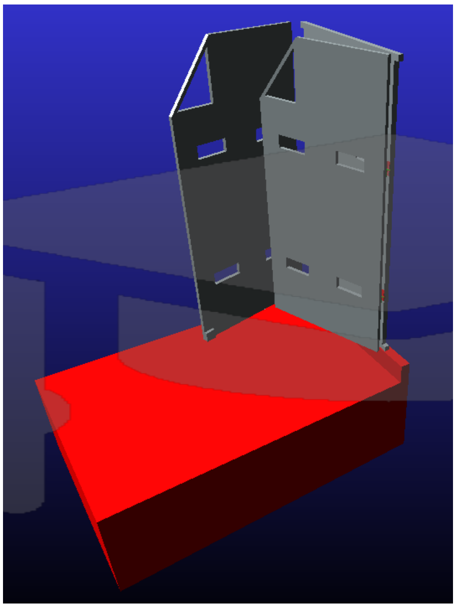

Figure 14: Opening of panels

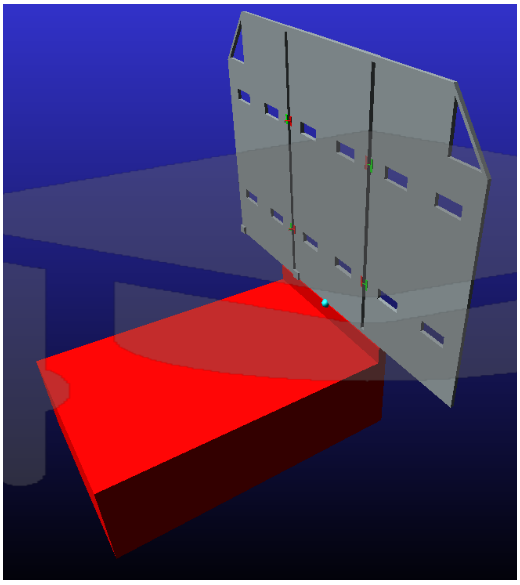

Figure 15: Panels fully opened

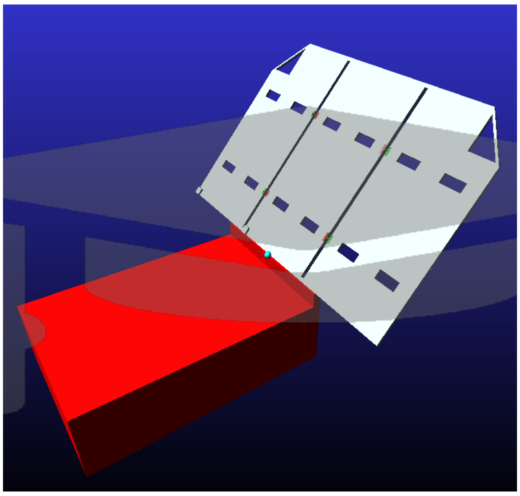

Figure 16: Sun tracking

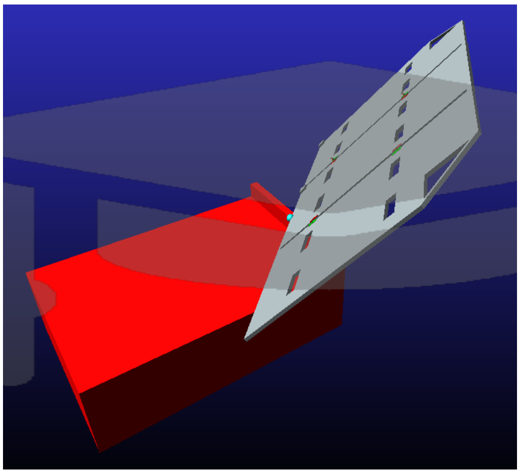

Figure 17: Sun tracking

2.5. Conclusion

It has turned out that modeling of the 3D multibody systems in Adams software is subjectively complicated and relatively time consuming. It is wise to fully develop the model in CAD software first, since any alterations of physical model are highly complicated in Adams. Even small detail like taking into consideration contact forces at the beginning of the process is worth thinking through. More often than not choosing appropriate markers for measurements of the model properties was not a trivial task. The process was however, very educational. Many problems had to be addressed: modeling joints, imposing motion functions to choosing realistic spring properties, just to name at few.
The end model consisted of thirteen bodies and fourteen connectors. Ten forces were controlled and the system was left with four degrees of freedom.

3. MATLAB Model

The objectives of the MATLAB models are to create a simplified version of the system to analyze its dynamics response during the deployment and to design a controller for the DC motor used in the first part of the above mentioned process.

Each of the three panels has been modeled as a rigid beam with mass m, length l. Each body is considered as independent from the others and is constrained to the ground using a *revolute hinge* at one of its extremities. As a consequence, the beam is only able to rotate around an axis passing through the hinge and perpendicular to the beam plane, with an inertia $J = \frac{1}{3}ml^2$.

3.1. Side panels deployment

Each side panel is connected to the ground also via a torsional spring with stiffness coefficient K computed as the sum of the stiffnesses of each of the springs attaching it to the middle panel and with a damper of coefficient C. The panels starts from an angular position of 0° and the springs present a prestrain of 180°. The joints parameters are the one sized using Adams and are summarized in Table 3.
The overall equations governing the motion of the two side panels are the one of a typical

spring-damper system:

$$\mathrm{M}\ddot{\boldsymbol{\vartheta}} + \mathrm{C}\dot{\boldsymbol{\vartheta}} + \mathrm{K}(\boldsymbol{\vartheta} - \boldsymbol{\vartheta_0}) = 0 \tag{1}$$

where $\boldsymbol{\vartheta}$ is the vector of the angular positions of the two panels, $\boldsymbol{\vartheta_0}$ is the vector of prestrain of the springs and M, C, K are the inertia, damping and stiffness matrices respectively.
In order to simulate the forces generated by the latches that prevents the panels from moving, two different strategies were implemented. Before the panels need to move, the prestrain of the corresponding spring is set to 0; once the panels reach the final position, their velocity and acceleration are simply set to 0.
Figure 18 shows the simulated positions and velocities of both side panels.

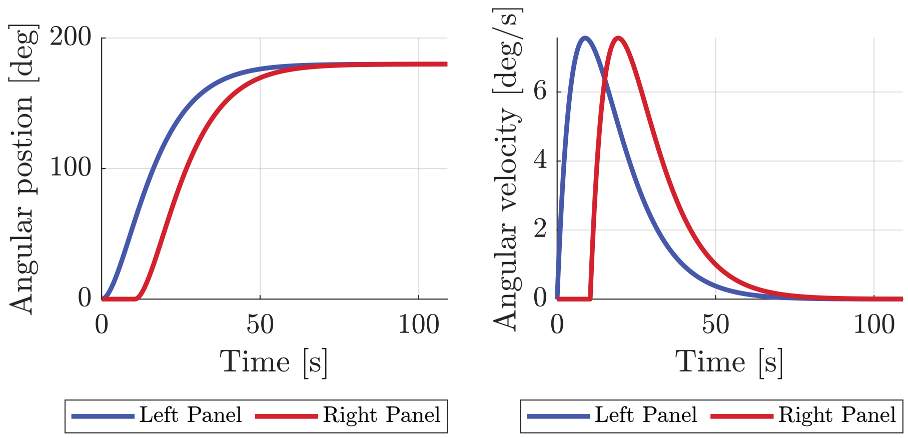

Figure 18: Angular position and velocity of the two side panels during deployment

3.2. DC motor model

As previously stated, the first part of the deployment requires an electric motor to rotate of 90 degrees the solar array assembly. Given the maximum torque of around 0.015 Nm found in the sizing process done in MSC Adams, the DB-1500 J-1ES brushless DC motor from MOOG, which guarantees a peak torque of 0.15 Nm, has been modeled [2]. The motor is mechanically characterized by a moment of inertia J_{motor} and is able to generate a torque proportional to the current I flowing in the armature coils, which are modeled as a resistance R and an inductance L in series. On the armature also acts a back electromotive force proportional to the angular velocity of the system.
The complete set of equations governing the motion of the coupled motor-solar array systems is represented in Equation 2.

$$\begin{cases} J_{sys} = J_{motor} + J_{array} \\ T = K_m I \\ \dot{\omega} = \dfrac{T}{J_{sys}} \\ \dot{\vartheta} = \omega \\ V = K_m \omega + RI + L\dot{I} \end{cases} \tag{2}$$

In order to effectively control the deployment, a simple cascade controller has been implemented.
A PI controller regulates the inner loop of the system, guaranteeing fast convergence of the

electric dynamics of the motor. It was designed neglecting the back EMF force and such that the open-loop system behaves as an integrator with an high cut-off frequency (500Hz).
The outer-loop controller is a PD controller designed assuming steady-state behaviour of the inner loop. It has a slow zero to guarantee stability and a very fast pole to ensure feasibility. The gain is designed such that the cut-off frequency of the open-loop system is 1Hz.

The block diagram of the overall system is shown in Figure 19.

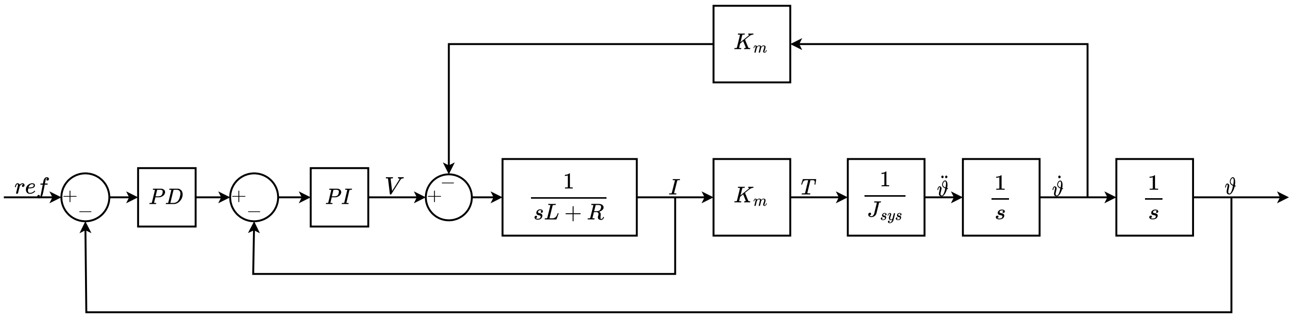

Figure 19: First deployment model's block diagram

As it has been done in MSC Adams, the reference position commanded to the system is a STEP5 function which goes from 0° to 90° in 300 seconds.

Figure 20 shows the evolution of the voltage and the current in the coils of the motor during the deployment. It can be seen that they are well below the rated limit of 4V and 15A respectively. Also the output torque is always much lower that the maximum 0.15Nm, as can be seen in Figure 21.
Figure 22 shows the angular position and velocity of the solar array during the deployment.

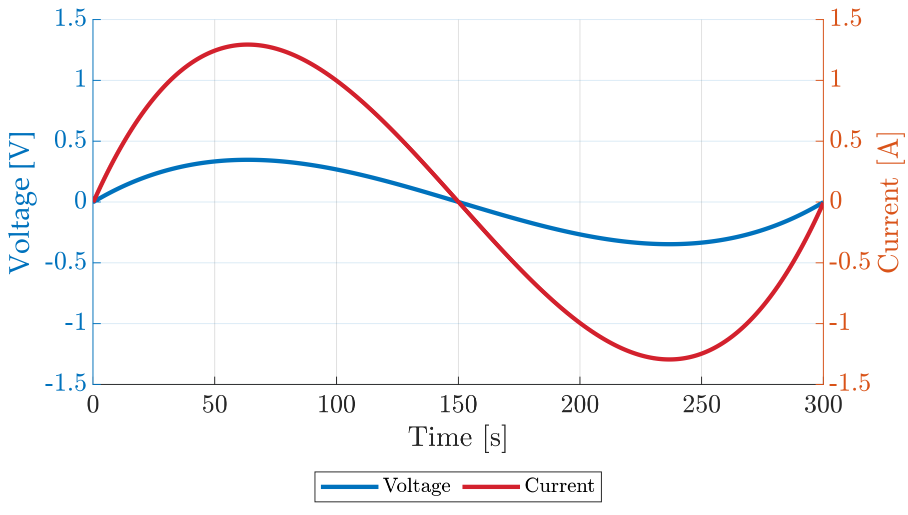

Figure 20: Current and Voltage on the DC motor during deployment

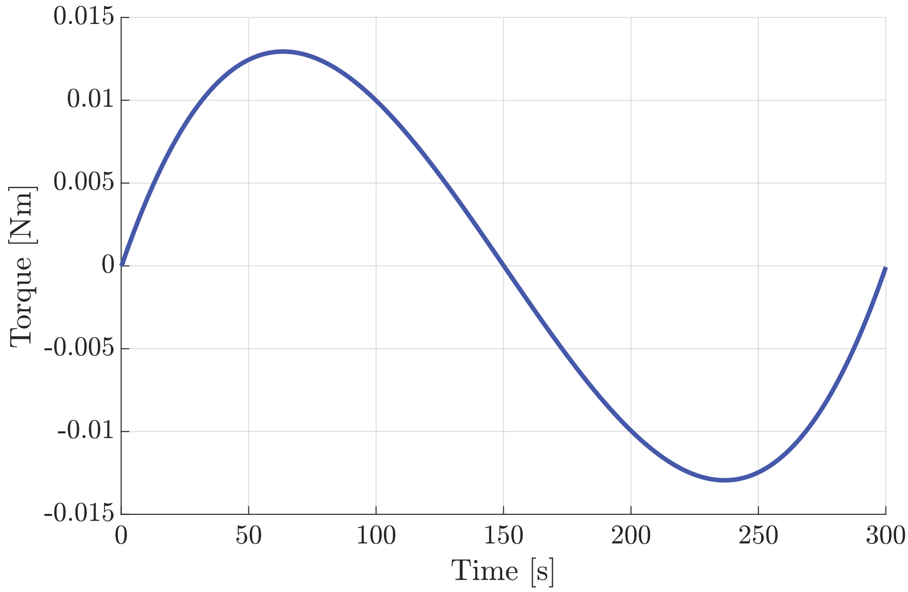

Figure 21: Torque generated by the DC motor during deployment

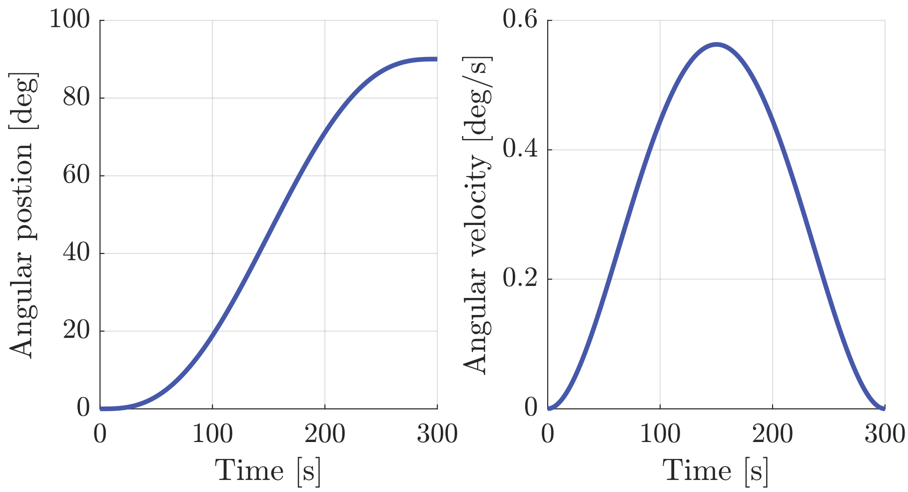

Figure 22: Angular position and velocity of the solar array assembly during deployment

3.3. Parameters generation for MBDyn

3.3.1 Input sequence

The reference position that the motor needs to follow in the MBDyn model has been approximated with a SISO model which best fitted the step5 input. The state-space matrices of the model were obtained using the MATLAB *tfest* function, which computes the transfer function that better fits the provided input-output data.
As it can be seen in Figure 32, the accuracy of the model is quite low for the first few seconds, where a significantly higher amount of torque is required, compared to the other models; after that the computed SISO model is able to follow accurately the step5 input, with some very low-amplitude and low-frequency oscillations once the final position is reached.

3.3.2 Motor parameters

Reproducing in MBDyn the model of the motor and its controller shown in Figure 19 is beyond the scope of this project; instead, the torque required to move the solar array can be generated

by an appropriate spring-damper system.
To find the correct values, the corresponding model of the simplified system has been implemented in simulink. Then the stiffness and damping coefficients have been optimized using the MATLAB built-in function *fmincon* in order to minimize the square of the difference between the torques provided in the two models. The equivalent stiffness and damping coefficient have been found to be $K = 1.722\,e5\ Nm/rad$ and $D = 2.791\,e4\ Nms/rad$ respectively. The results of such optimization are shown in Figure 23. It can be seen that both the torque and the solar array position in the approximated case are almost identical to their corresponding values obtained using the more detailed model.

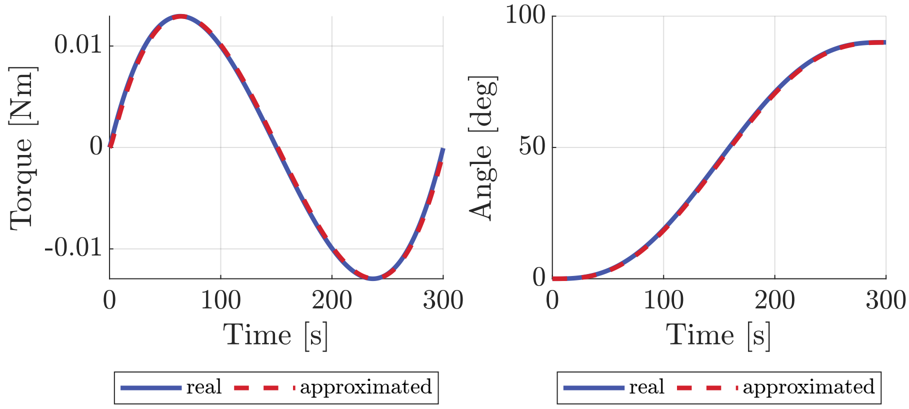

Figure 23: Comparison of torque and angular position between the real system and approximated one

4. MBDyn model

4.1. General Overview

The previously depicted compound, was further simplified to be implemented in MBDyn. In particular the panels were considered as completely rigid and a first analysis was conduced grounding the whole system. In the attempt of sticking to reality as much as possible, the DC motor was modeled through a third order SISO system, while four different springs and two dampers were used to connect the three bodies. Here an initial value problem was solved by means of a multistep integration method, setting tolerances on the derivatives and convergence criteria in the order of 10^{-6}.

4.2. Nodes

Deepening into the model, a series of six nodes was used to individuate the relevant points of the problem. In particular only the three related to the panels were chosen as *dynamic* in the *structural* category, to have inertia properties of the bodies associated to them, while the ones individuating the ground and the actuator were selected as *static*, since no inertia was related to those and therefore saving six degrees of freedom. An additional one individuated as *abstract*, since it only contains the output of the SISO model (used as the imposed motion of the dc motor), was used.

4.3. Reference Frames

A reference frame was attached to each node providing an orientation in space for these. The grounded one was simply oriented as the system of coordinate indicated as *global* in the software, while all the others referring respectively to the *ACTUATOR* and the the *PANELS* were rotated for commodity reasons using the Euler angles of -90° around the global *x-axis*. As will be explained in the following paragraphs an extensive use of the *deformable axial joint* will be considered; since one of its characteristic is that of allowing for a rotation only around the *z-axis* of the reference frame in which the joint is defined, a smart definition of these can result as extremely useful, avoiding continuous rotations.

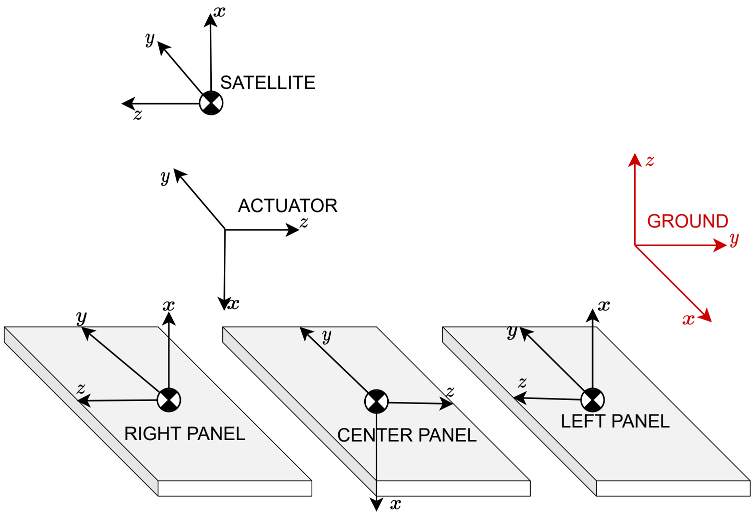

Figure 24: Reference Frames

4.4. Elements

Three *body elements* were attached to the structural dynamic nodes individuating the panels. Mass and inertia properties were therefore specified as illustrated in Appendix A.

As explained in subsection 3.3, the input of the DC motor was reconstructed using a state space SISO of third order which, coupled with a deformable axial joint, drives the position of the the main assembly during the first act of the deployment. The SISO is individuated in the code as *genel* (standing for generic element) whose output is stored by means of a *drivecaller* into the abstract node previously mentioned.

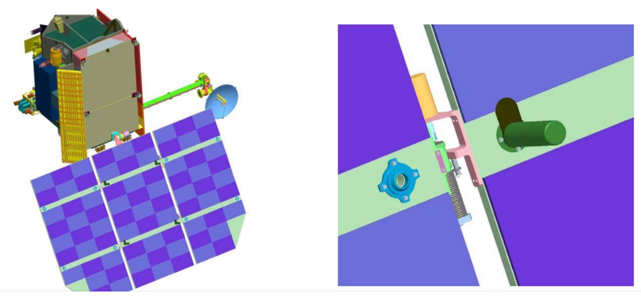

Figure 25: Hinges details

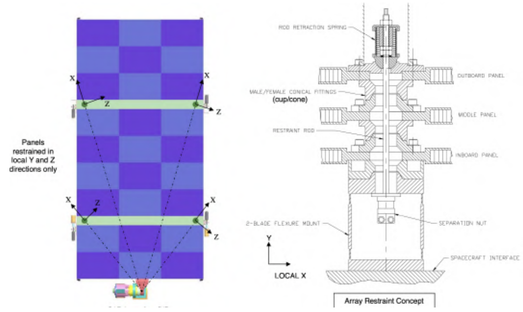

Figure 26: Hinges' details

A set of twelve joints were in the end used in order to correctly constraint all the nodes in a proper way. Starting from *NODE GROUND*, this was fixed through a clamp constraining all its degrees of freedom. Two more linking systems were added so to attach the *ACTUATOR* node to the *GROUND* one. In particular a *total joint* allowing only for the rotation around the *z-axis* in the *ACTUATOR reference frame*, was used. Overlapped to the latter, a *deformable axial joint* was inserted to simulate through an equivalent spring-damper system, prestrained by means of the above mentioned motor input, the rotation of the assembly. The degrees of freedom of the middle panel were fully constrained to the *actuator node* through a full active *total joint*, making these two to act and therefore move as a whole.

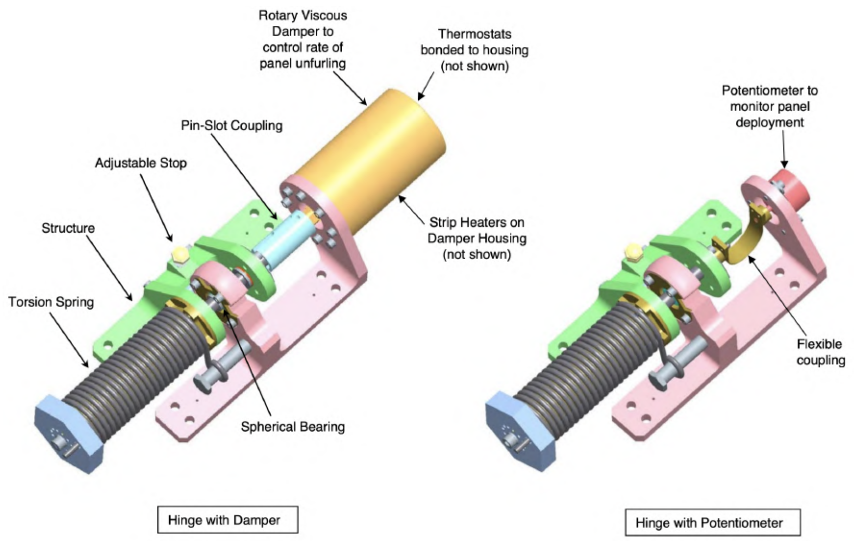

Figure 27: Panel to panel real hinges

Figure 27 shows the real hinges mounted in the side to side panel connections. These consist of a double mechanisms presenting a prestrained spring in both cases to which a damper in one and a potentiometer in the other are attached. In the attempt of sticking with reality as much as possible, this double connection was replicated in the code using four *total joints* overlapped with an equal number of *deformable axial joints* among which two were equipped with dampers. A series of *bistops* governing the activation and deactivation of these joints were used to allow for the correct deployment sequence. In particular the first opening is the left-hand one, therefore a law is used to make this happen only once the whole assembly reached the 90° of deployment. The one on the right side instead starts opening once the left one reached the 60° of aperture. All the springs are prestrained within an angle of 180°, which together with the bistop contributes to correctly arrest the panels in their motion once these are fully deployed.

4.5. Results

As anticipated in the introduction, the goal was to use MBDyn to analyze the movements of the system developed and the torque required, once the motor and the spring-damper mechanism of the panel to panel linkers were sized through Matlab and Adams.
The deployment sequence consisted of two independent movements, differently governed. The first one, guided through the DC motor, is aimed at bringing the whole assembly to a 90° configuration with respect to the satellite body, to which the panels were laid on. As shown on Figure 28, the actuation system is sized to end this first act of the deployment in 300s. Because of a required initial acceleration different from zero, an initial peak in the angular velocity can

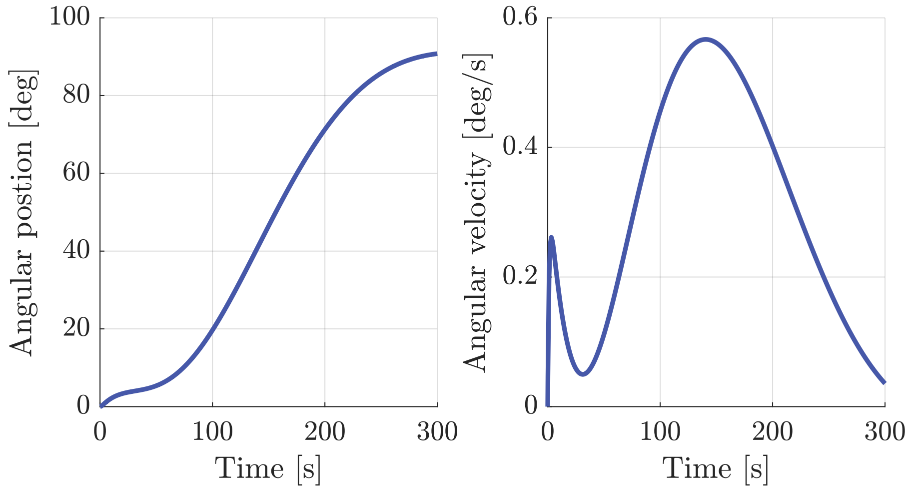

Figure 28: First Deployment

be seen. The second peak is due to a progressive increment of the deployment velocity which is then slowed down once the system starts approaching the final position. Confirmations of previous statements are found in the image displaying the behavior of the angular position where a curvature change is evident in the first 20 seconds of the simulation (impulsive start followed by the deceleration), while a the position tend to stabilize around 90°, due to the angular velocity of the actuator being brought to zero.
For what concerns the lateral panels' deployment, similar considerations can be done. The body initiating the opening sequence is that placed on the left, looking at the system from the satellite body. The input is as impulsive as that given by a precharged spring can be. Therefore a sudden acceleration is experienced, bringing the angular velocity to reach is maximum in the first 20 seconds of the simulation; as a consequence great part of the movement occurs in less that 50 seconds after which a flattening behaviour of the angular position curve, caused by the dampers, is displayed. The right panel is set into motion when the left one reaches 60° in the opening process and shows an identical behaviour in terms of angular position and rate (as expected being the two bodies equal as well as the joints system).

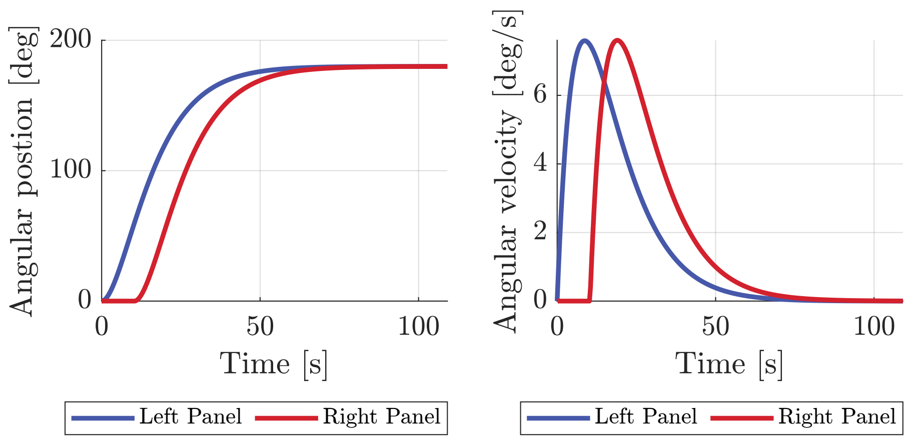

Figure 29: Second Deployment

The behaviour of the reactant torque at the actuator, exactly resembles what was shown in the second graph of Figure 28. The impulsive input of the motor results in a counter couple presenting a negative peak towards the very beginning of the deployment sequence. It is worth noticing that during the two deceleration processes a negative concavity of the curve is shown while a positive one follows the speed increasing mechanism. However being the movement slow and particularly smooth, after the first peaks, just what seems to be small oscillations around the zero are manifested, but they match the couple obtained through the motor sizing.

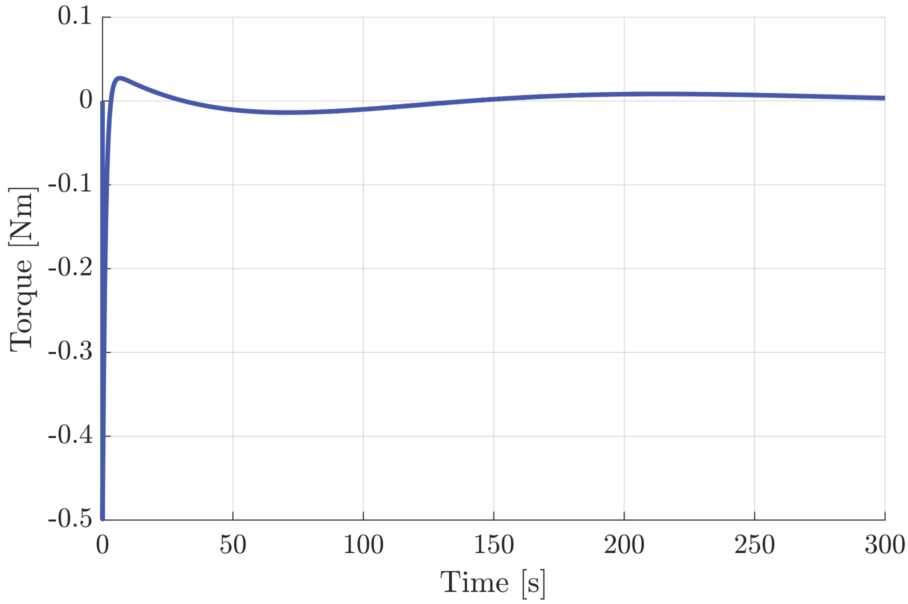

Figure 30: Torque at the actuator

4.6. Satellite

A further development of the previous model included the substitution of the grounded node with a body representing that of the satellite, left free to move in space. No substantial changes were made to the above explained construction, but for fixing the ACTUATOR to the SATELLITE in a way that only the rotation about the *z-axis* was allowed (in the actuator reference frame) as was done in the previous system with respect to the ground node. The SATELLITE itself was considered as a simple *element:body* attached to a *structural dynamic node*, for which inertia properties and mass were specified according to the information found in literature.
Hereafter, Figure 31 displays the rotation angles of the whole assembly (which is solidly moving with the satellite body). These are computed with respect to the global reference frame, neglecting the initial rotation between this coordinate system and that attached to the satellite body. As expected, due to the inertia of the panels, during both their opening motions, some minor displacements are followed as response by the system. In particular, rotating the whole solar array causes a significant rotation of the satellite's main body around the same axis (yellow curve). The imbalance generated by the asymmetric deploying of the side panels (one after the other) is clearly visible also (peaks). However the overall effect of the side panels opening cancels out once they both converge to the 180°, resulting in a negligible rotation around the x and y axes of the ground reference frame.
The movements are not changing the spatial arrangement of the assembly in terms of translations; this happens thanks to the huge mass of the satellite if compared to that of the panels and to the small displacements of the baricenter of the whole system.

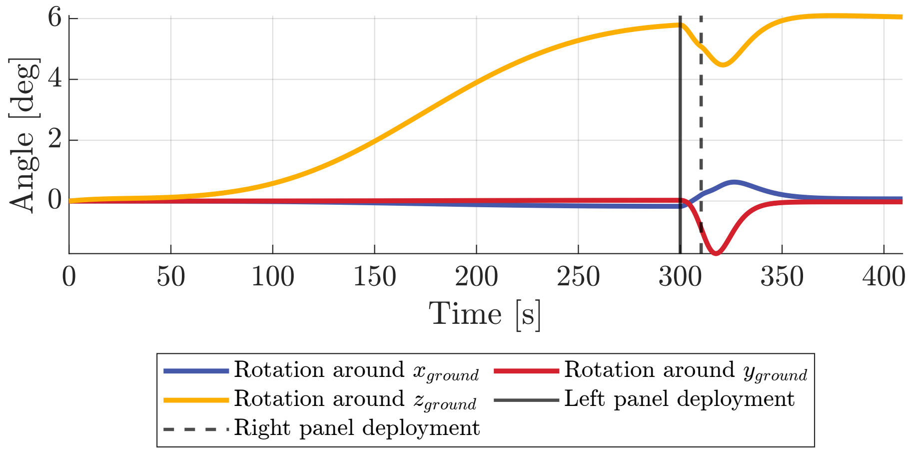

Figure 31: Satellite rotations around the ground reference system

5. Results Comparison

All the results previously presented are here gathered together. As expected, having followed the exact same procedure in Adams and Matlab, very minor discrepancies are shown among the two models, while a significant difference is shown at the beginning of the simulation comparing these two with the MBDyn one. Reason of such behaviour lays in the way the motion of the actuator is driven. The different commanded position causes an impulsive start which reflects in the peaks on the torque and angular velocities.

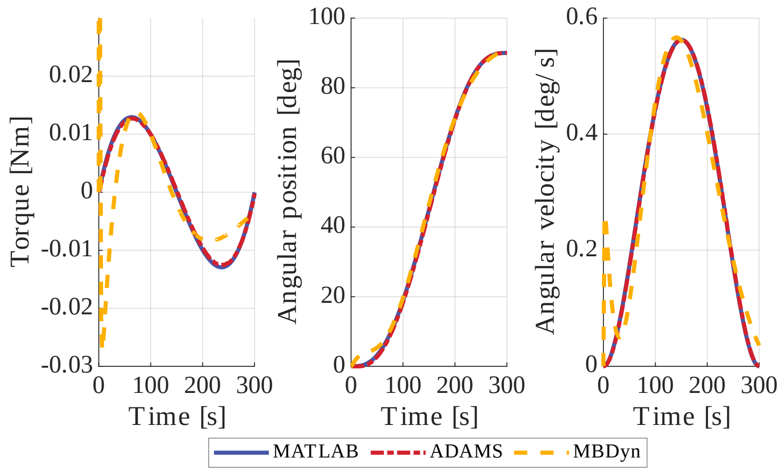

Figure 32: First Deployment

For what concerns instead the lateral deployment, no significant differences are shown. The precharged spring-damper system behaves exactly in the same way in the three models. No dynamic modelling of particular subsystems is here involved therefore there is no reason for the physics to be different in the three cases.

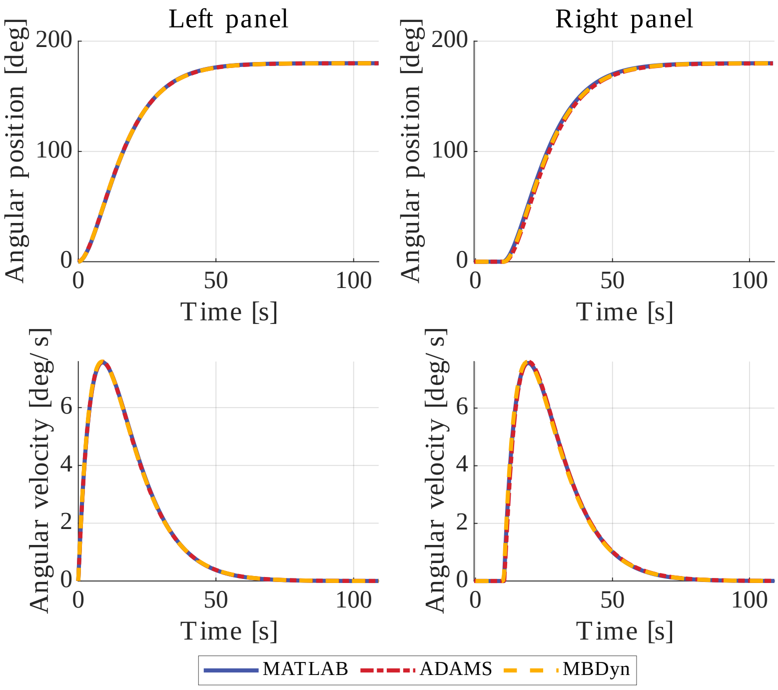

Figure 33: Second Deployment

A. System Data

Satellite			
Satellite Mass	1853	kg	
Satellite Inertia	$\begin{bmatrix} 677.8 & 69.8 & -6.2 \\ 69.8 & 1038.6 & -32.5 \\ -6.2 & -32.5 & 1192.9 \end{bmatrix}$	$kg \cdot m^2$	ref. frame: see Figure 1, Figure 24

Table 1: Satellite properties

Solar Array			
Center Panel Mass	20.208	kg	
Side Panels Mass	14.451	kg	
Center Panel Inertia	$\begin{bmatrix} 16.616 & 0 & 0 \\ 0 & 13.507 & 0 \\ 0 & 0 & 3.118 \end{bmatrix}$	$kg \cdot m^2$	ref. frames: see Figure 24
Side Panels Inertia	$\begin{bmatrix} 11.079 & 0 & 0 \\ 0 & 8.989 & 0 \\ 0 & 0 & 2.093 \end{bmatrix}$	$kg \cdot m^2$	
Panels length	2.8	m	
Panels width	1.347	m	

Table 2: Solar Array properties

Panels Joints		
No. of Joints	2 per side panel	
No. of Springs	2 per side panel	
No. of Dampers	1 per side panel	
Springs Stiffness	1e-3	Nm/rad
Springs pre-strain	180	deg
Damping Coefficient of the Damper	2	Nms/rad

Table 3: Joints properties

DC Motor		
Peak Torque	1.5	Nm
Motor Constant	0.01	Nm/A
Resistance	0.268	Ohm
Inductance	0.076	mH
Inertia	0.064e-5	$kg \cdot m^2$

Table 4: DC motor parameters

Controller gains			
PI		**PD**	
$P + I\frac{1}{s}$		$P + D\frac{N}{1 + N\frac{1}{s}}$	
P	0.239	P	59.81
I	841.9	D	565.55
		N	62.81

Table 5: Controller gains

References

[1] A. Bartels, M. Reden, L. Hartz, J. Baker, S. Scott, and T. Jones, "LRO preliminary design review," Tech. Rep., 2006.

[2] MOOG. Db-1500 matrix motor technical datasheet. [Online]. Available: https://www.moog.com/content/dam/moog/literature/MCG/DB-1500.pdf